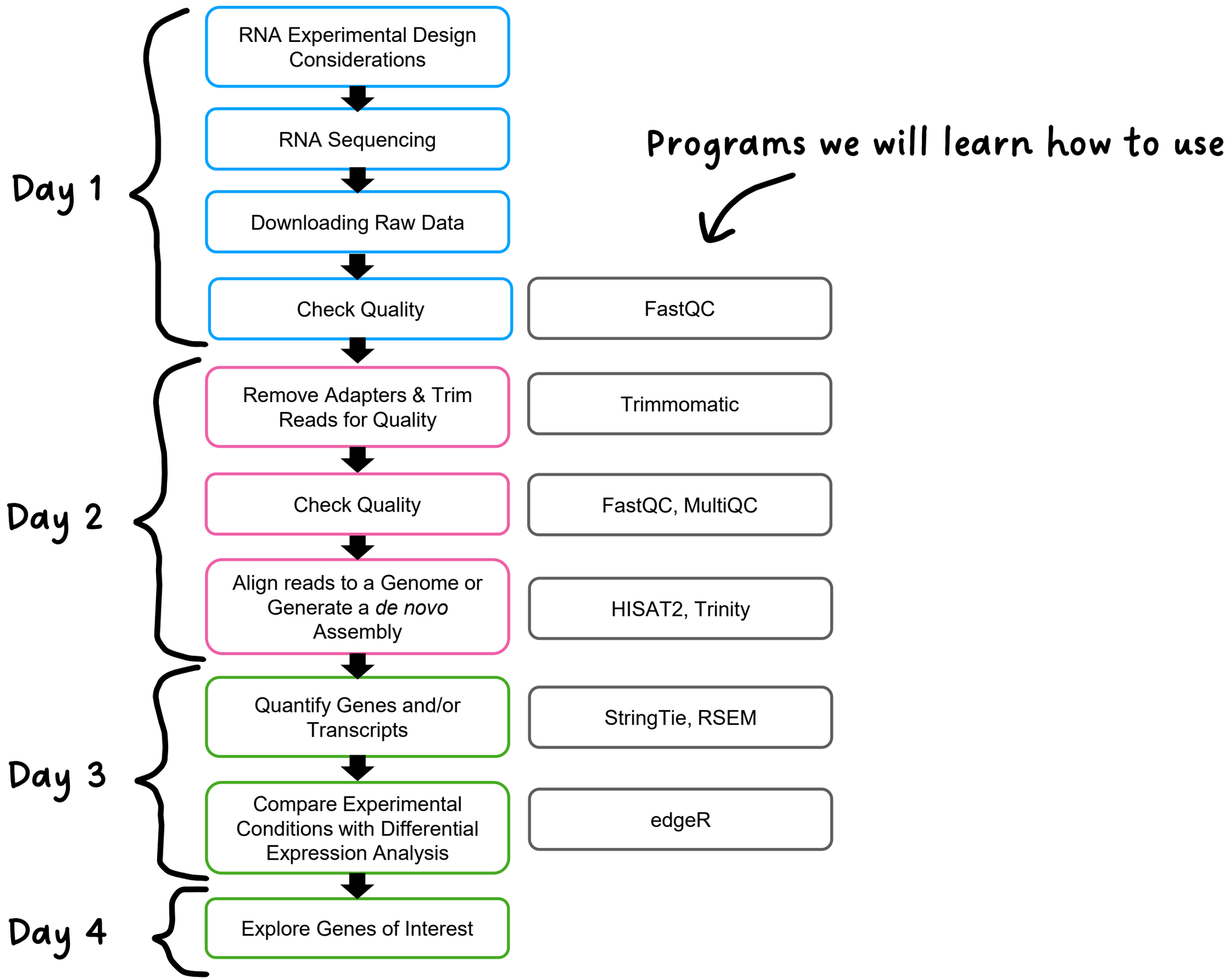
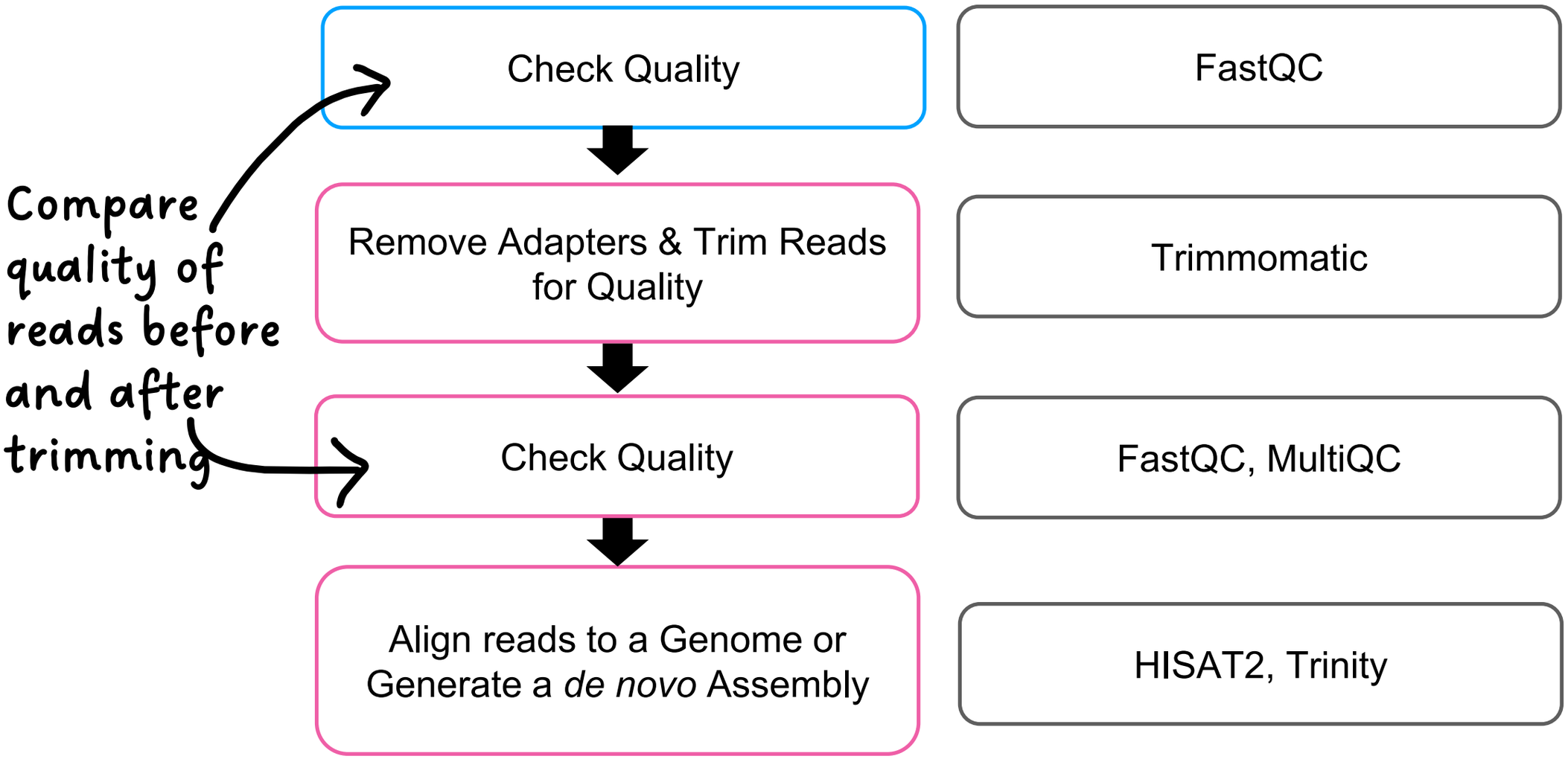
RNA-seq and analysis of differential gene expression

Slides adapted from: Tom Booker & Bernadette Johnson
Bioinformatics for Evolutionary Biology and Conservation

Basic RNA-seq pipeline



Basic RNA-seq pipeline



Assessing Quality & Trimming

- How can I check the quality of my reads?
- What do I need to trim out of my data?
- How do I trim my data?

Assessing Quality & Trimming

Last time we used:

- FastQC

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Input: fastq file → **Output: results.html**

- per base sequence quality
- per sequence quality score
- overrepresented sequences
- adapter content

Last time we ran FastQC, now we can check the quality report of our raw reads!

FastQC html report

FastQC Report

Summary

- Basic Statistics
- Per base sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Stats tells you if everything went according to plan when sequencing

Basic Statistics

Measure	Value
Filename	SRR19736866_1.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	17081609
Sequences flagged as poor quality	0
Sequence length	150
%GC	45

close to sequencing depth requested ?

read length correct for sequencing strategy?

GC content expected for my species?
(for eukaryotes its ~40-50%, for bacteria and archaea it can vary ~20-70%)

FastQC html report

This report is important!

FastQC Report

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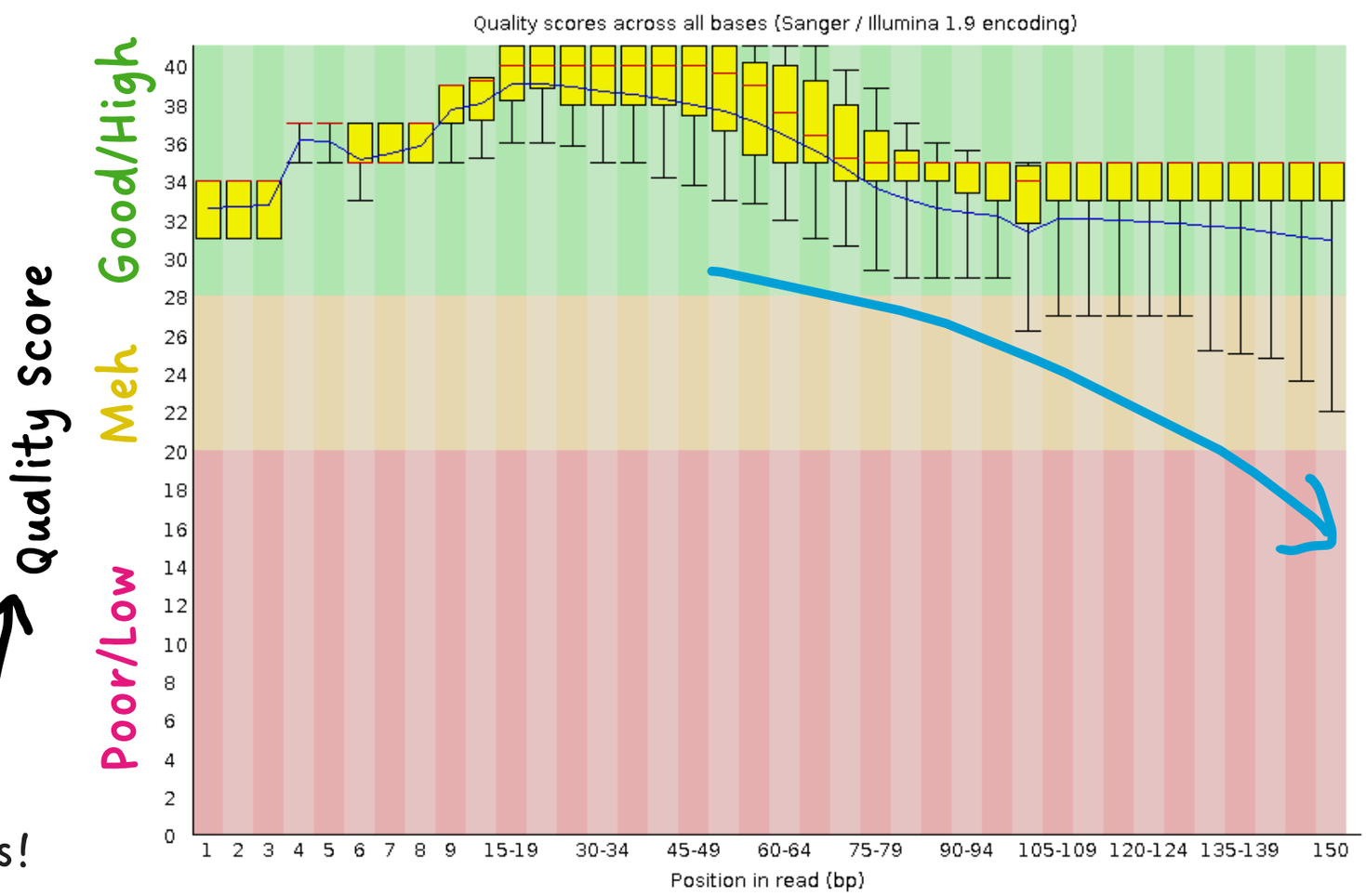
Per base sequence quality



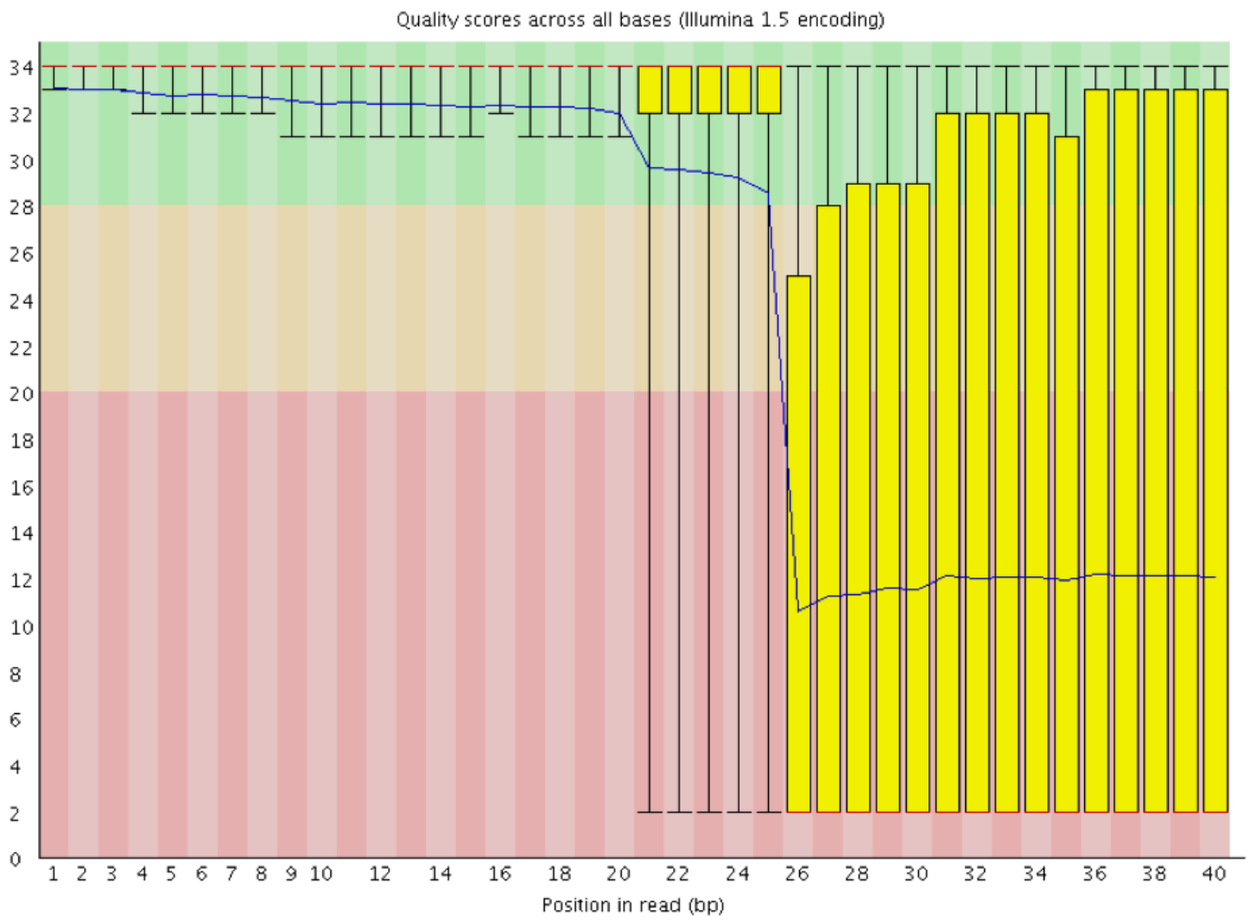
usually will see signal decay at the 3' end (so some drop is expected)

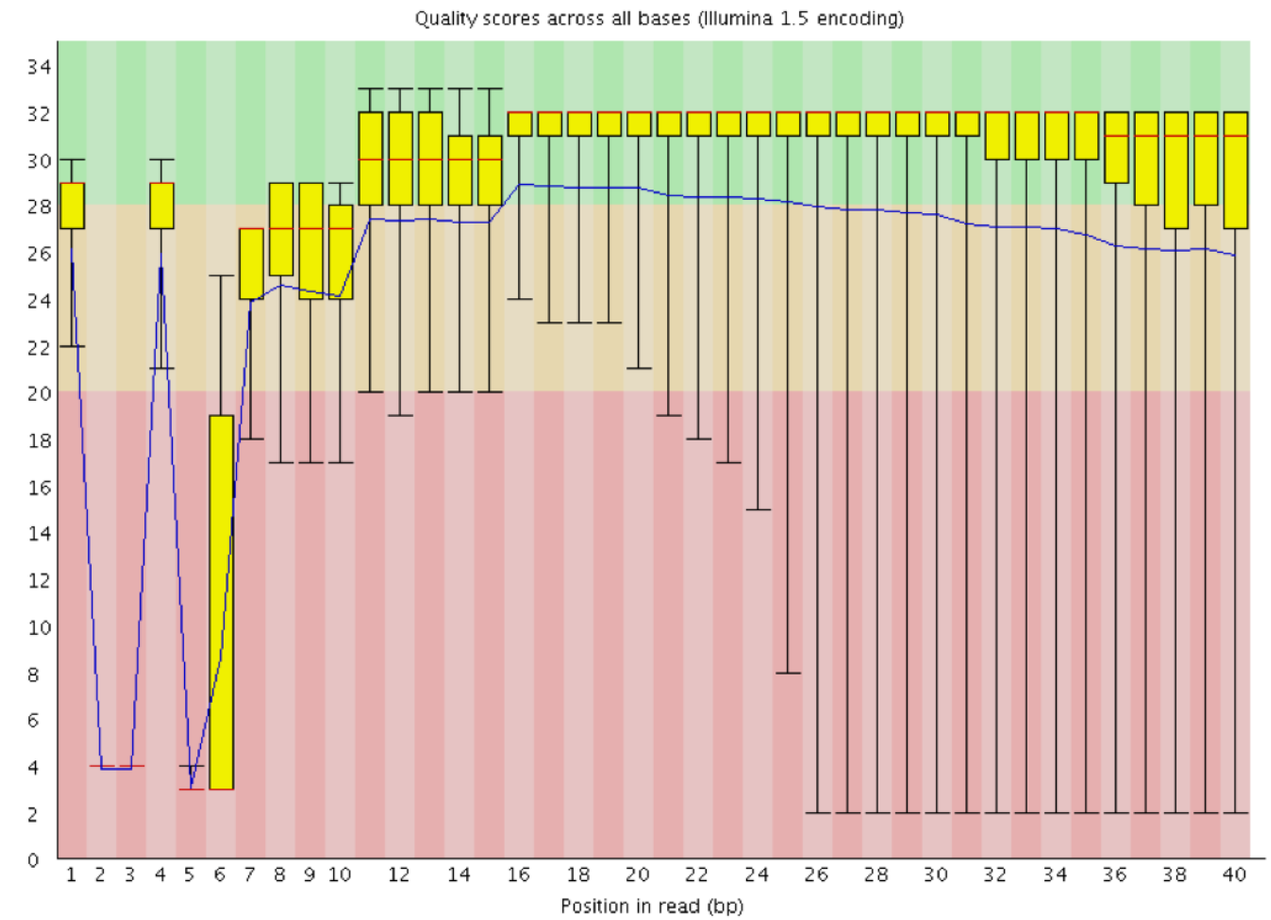
These are phred scores!
-20 means there is a 1 in 100 chance of an incorrect base call (or an accuracy of 99%)
- 30 = 99.9%
- 40 = 99.99%

the quality of the nucleotide base calls are related to the signal intensity and purity of the fluorescent signal

FastQC html report

Our sample looks great, but major issues might look like this:





If you have this, you should contact the sequencing facility!

If you have lots of reads, and you trim away the poor quality, you might have enough reads left to save the data.

https://hbctraining.github.io/Intro-to-rnaseq-hpc-salmon/lessons/qc_fastqc_assessment.html

FastQC html report

This one is also important!

FastQC Report

Summary

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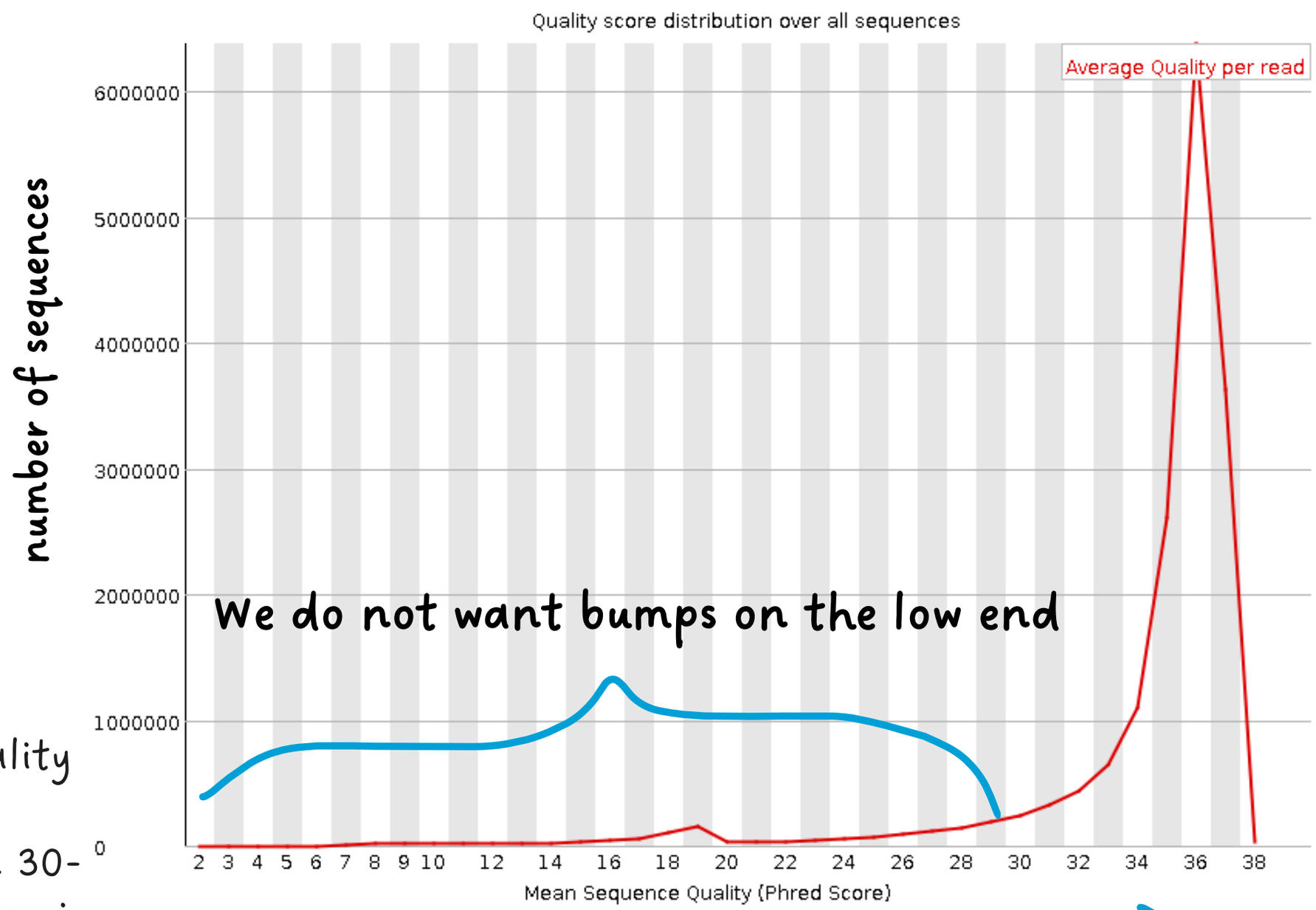
Per sequence quality scores



Our average sequence quality is great!
Remember scores between 30-40 mean base call accuracy is between 99.9%-99.99%

We want our reads to have high quality

FastQC html report

This report will usually fail for RNA-seq data!

FastQC Report

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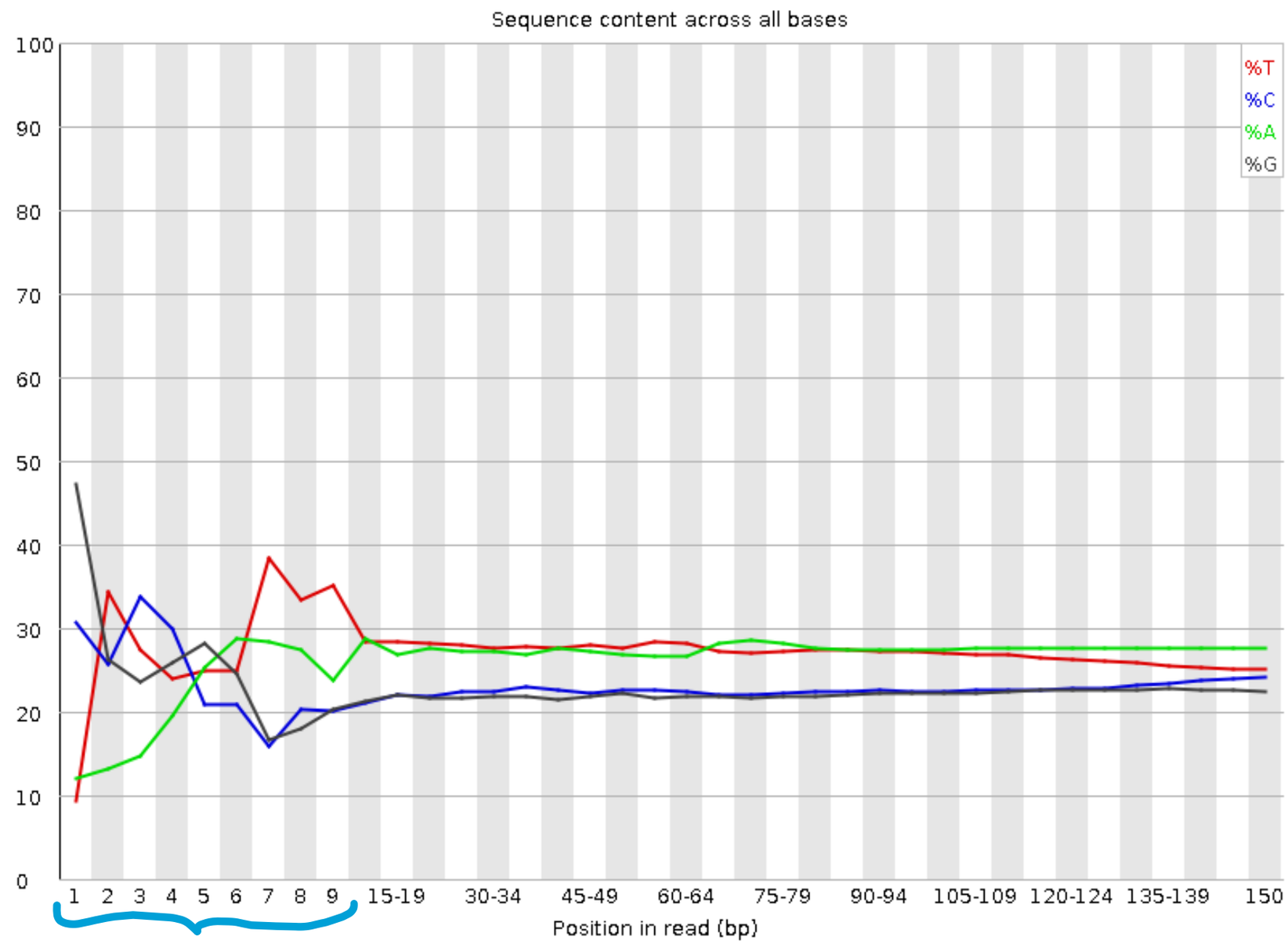
Per base sequence content



The first 10-12 bases result from the 'random' hexamer priming that occurs during RNA-seq library preparation.

FastQC html report

FastQC Report

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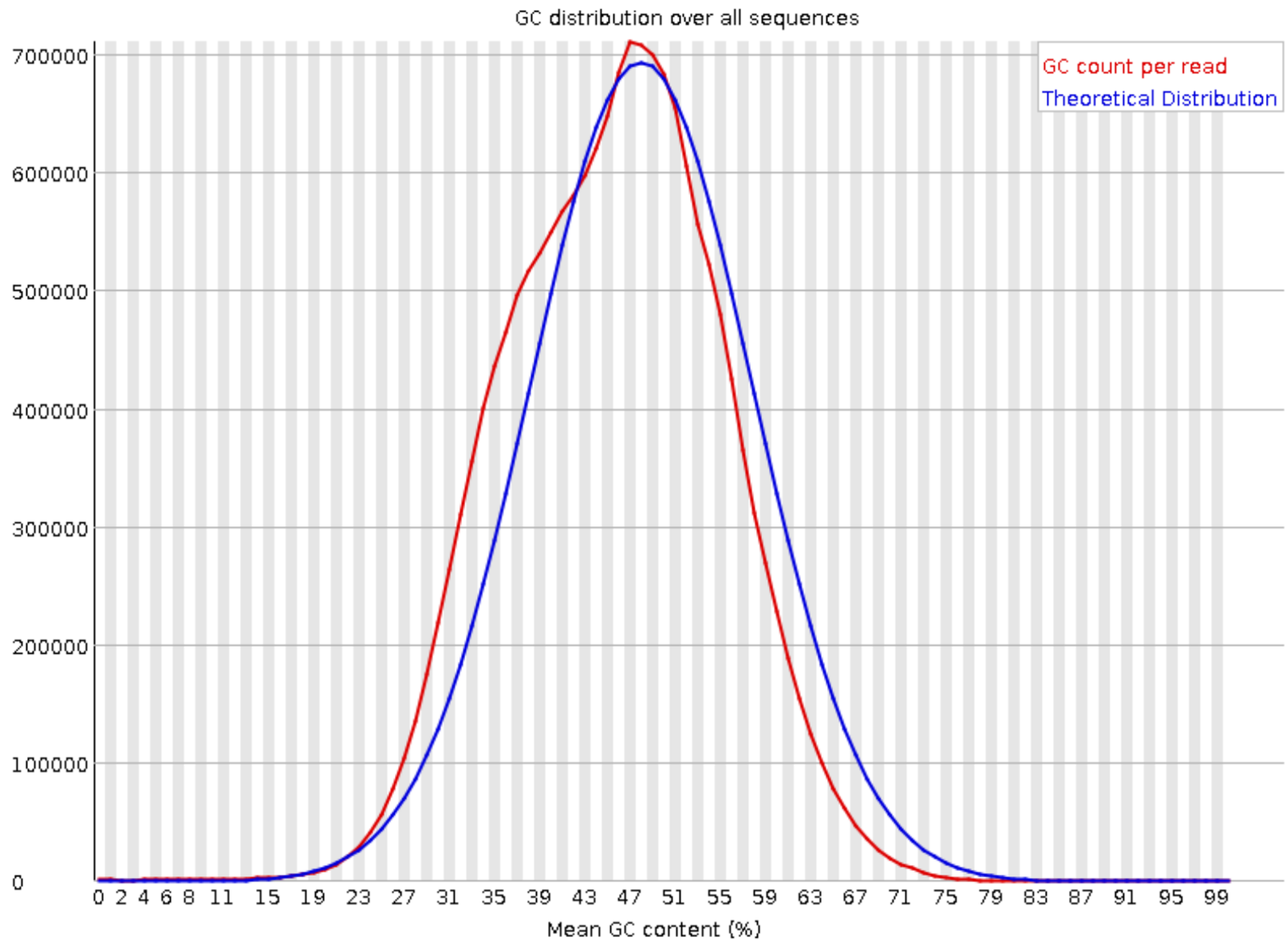
Per sequence GC content



This one is not as important, but worth checking.
-broad peak = possible contamination with another organism
-sharp peak = possible overrepresented sequences

FastQC html report

FastQC Report

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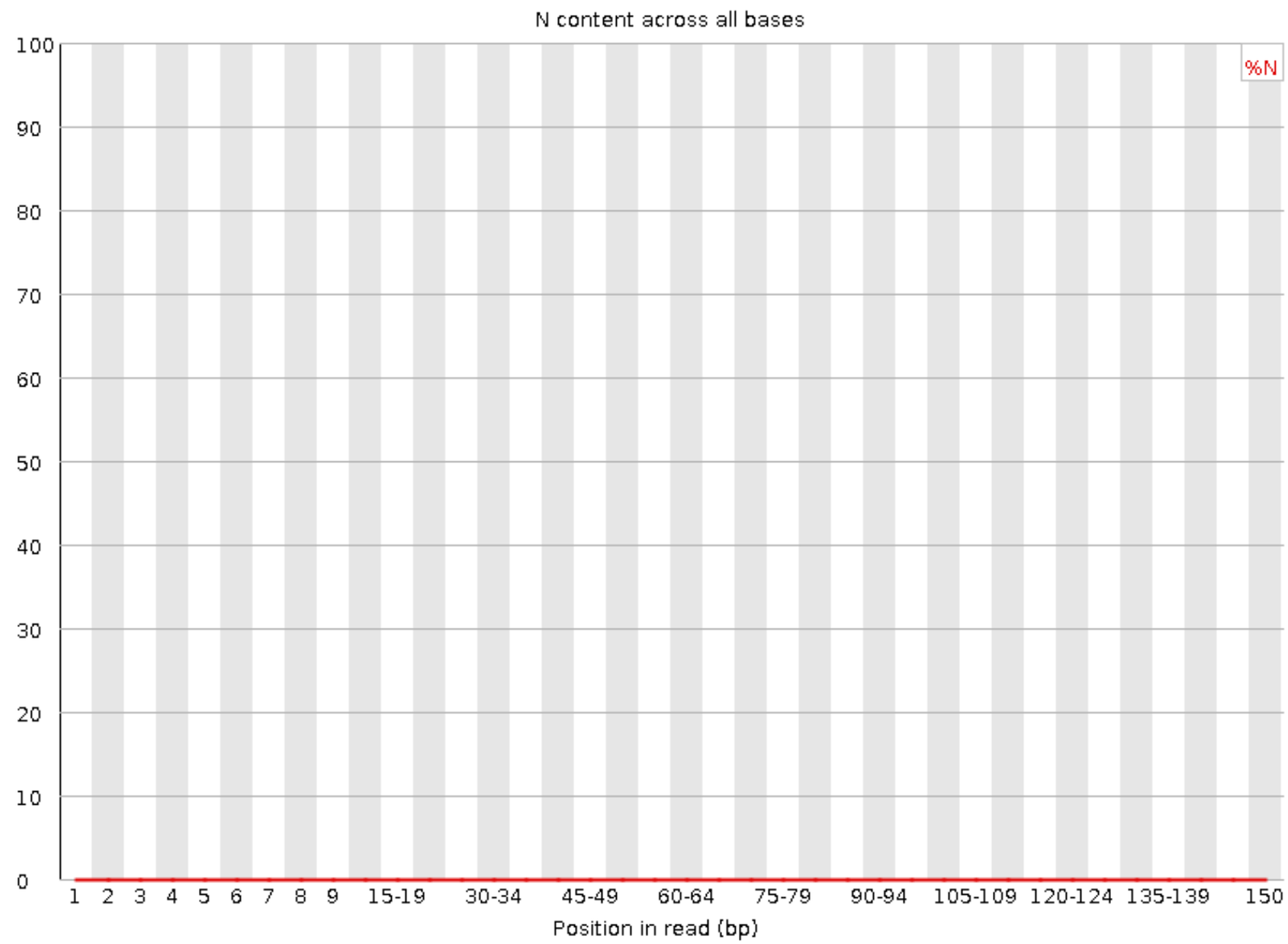
Per base N content



This shows the percentage of bases at each position or bin with no base call.
- peaks above 0 (anywhere) = problem during sequencing

FastQC html report

FastQC Report

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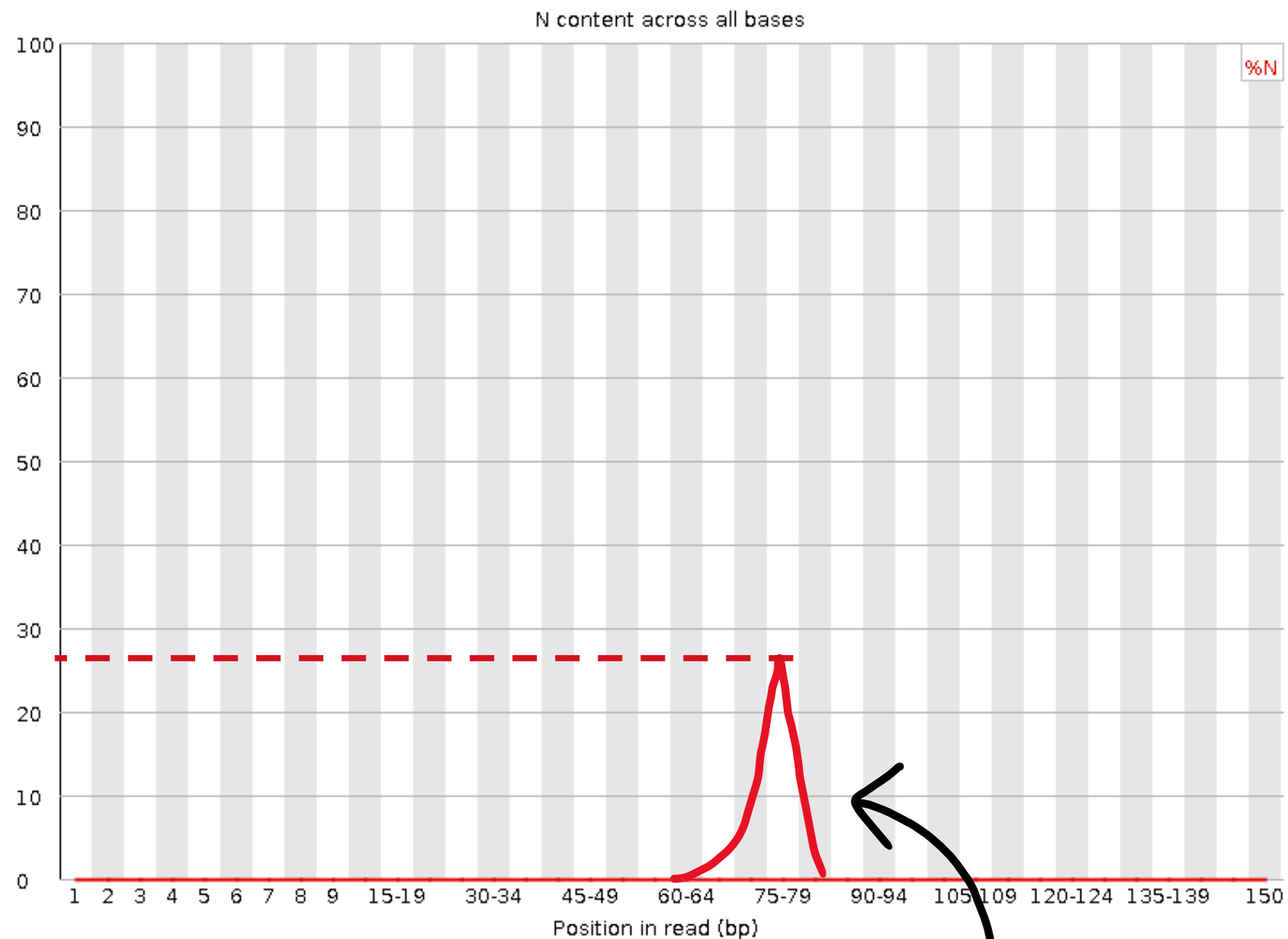
Per base N content



A peak like this would mean that the sequencer could not call a base for ~28% of reads at position 75-79.

FastQC html report

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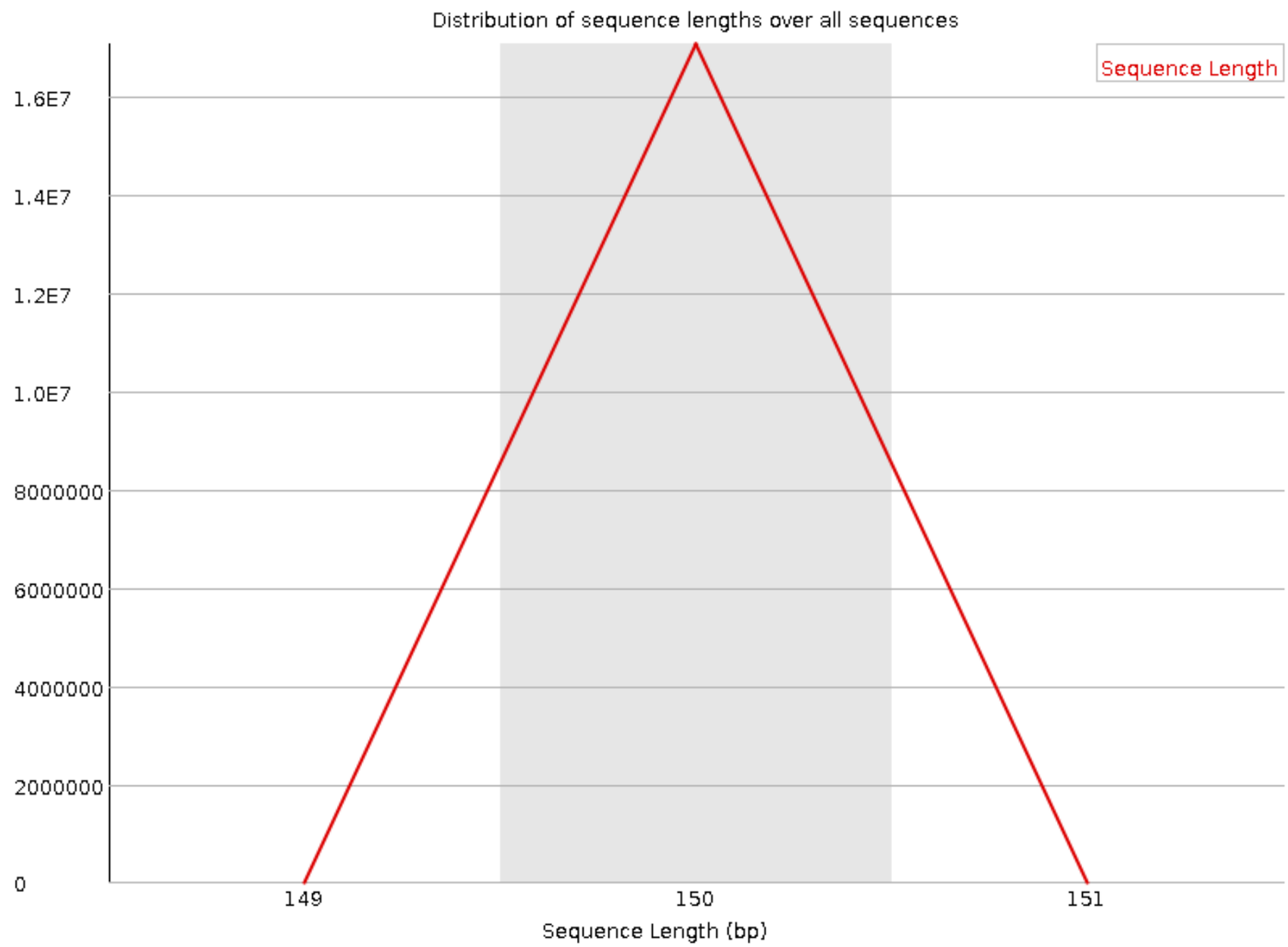
Sequence Length Distribution



This shows the distribution of sequence lengths. Any deviations from the expected bp count would indicate an issue with the library prep step.

FastQC html report

RNA-seq data usually fails this!

FastQC Report

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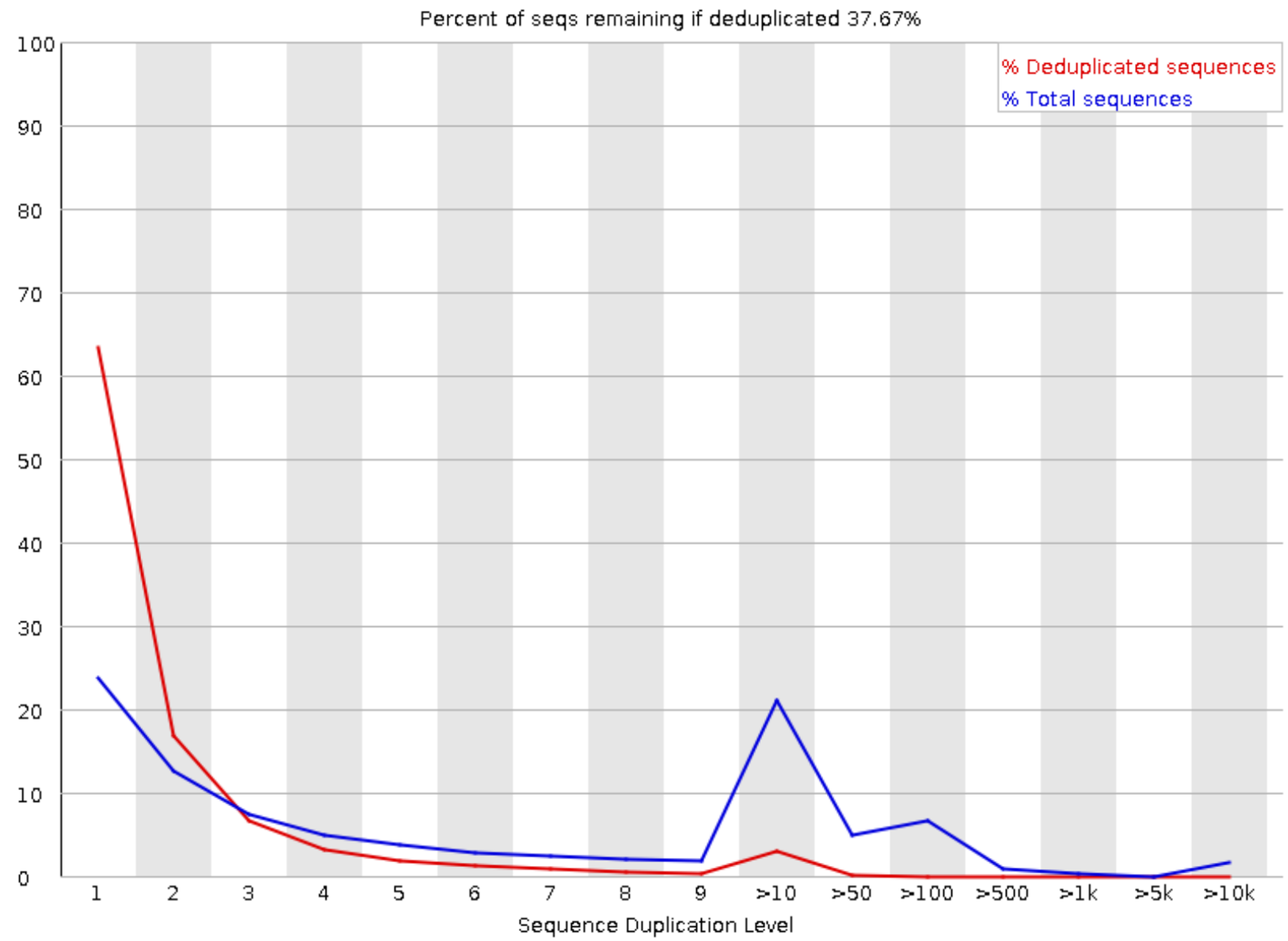
Sequence Duplication Levels



For this we are going to focus on the blue line. This represents the percentage of reads of a given sequence in the sample, which are present a given number of times.

FastQC html report

FastQC Report

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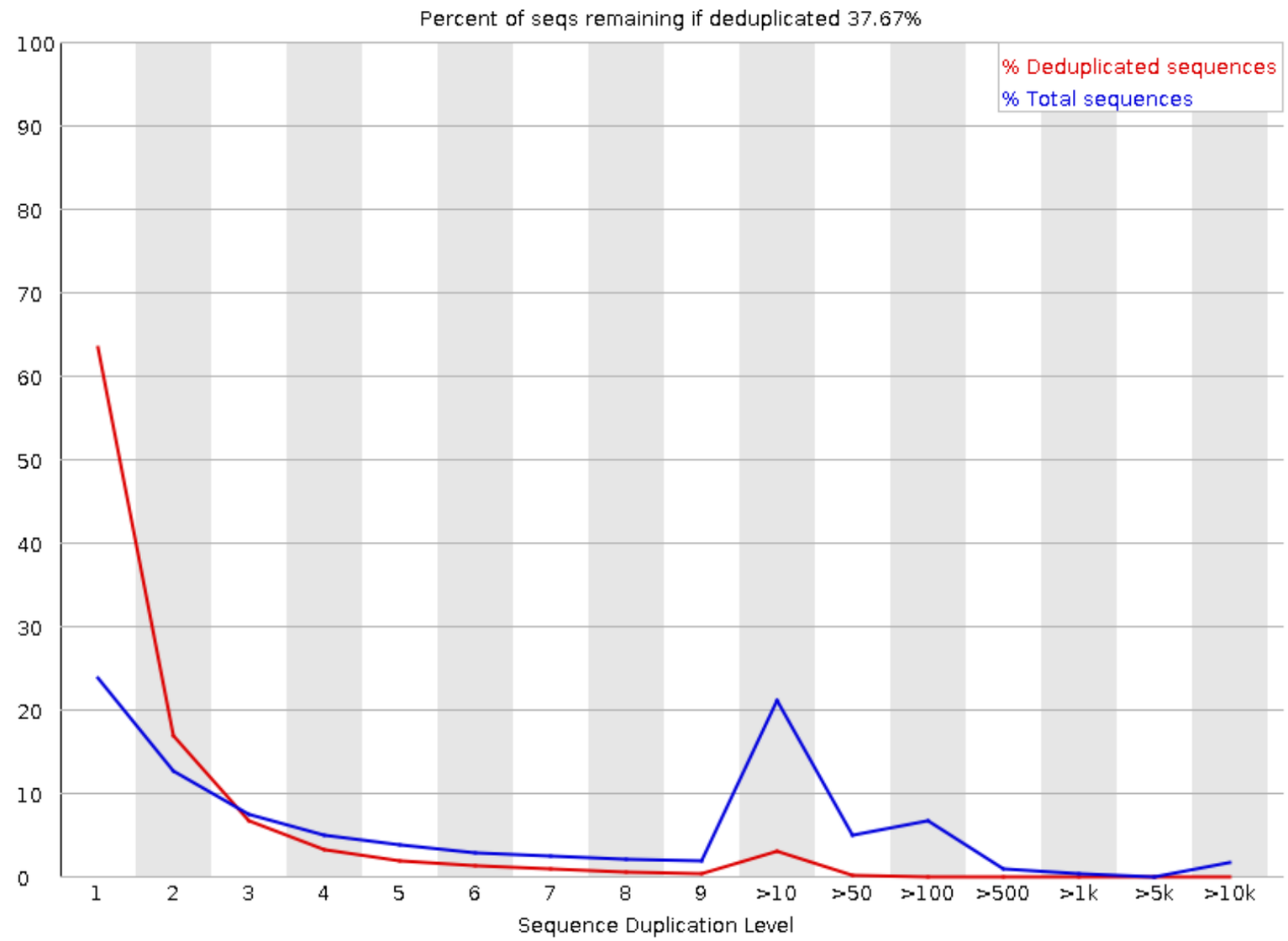
Sequence Duplication Levels



This can be due to PCR overamplification, or over-represented sequences (highly expressed sequences).
How can we tell?

FastQC html report

FastQC Report

Summary

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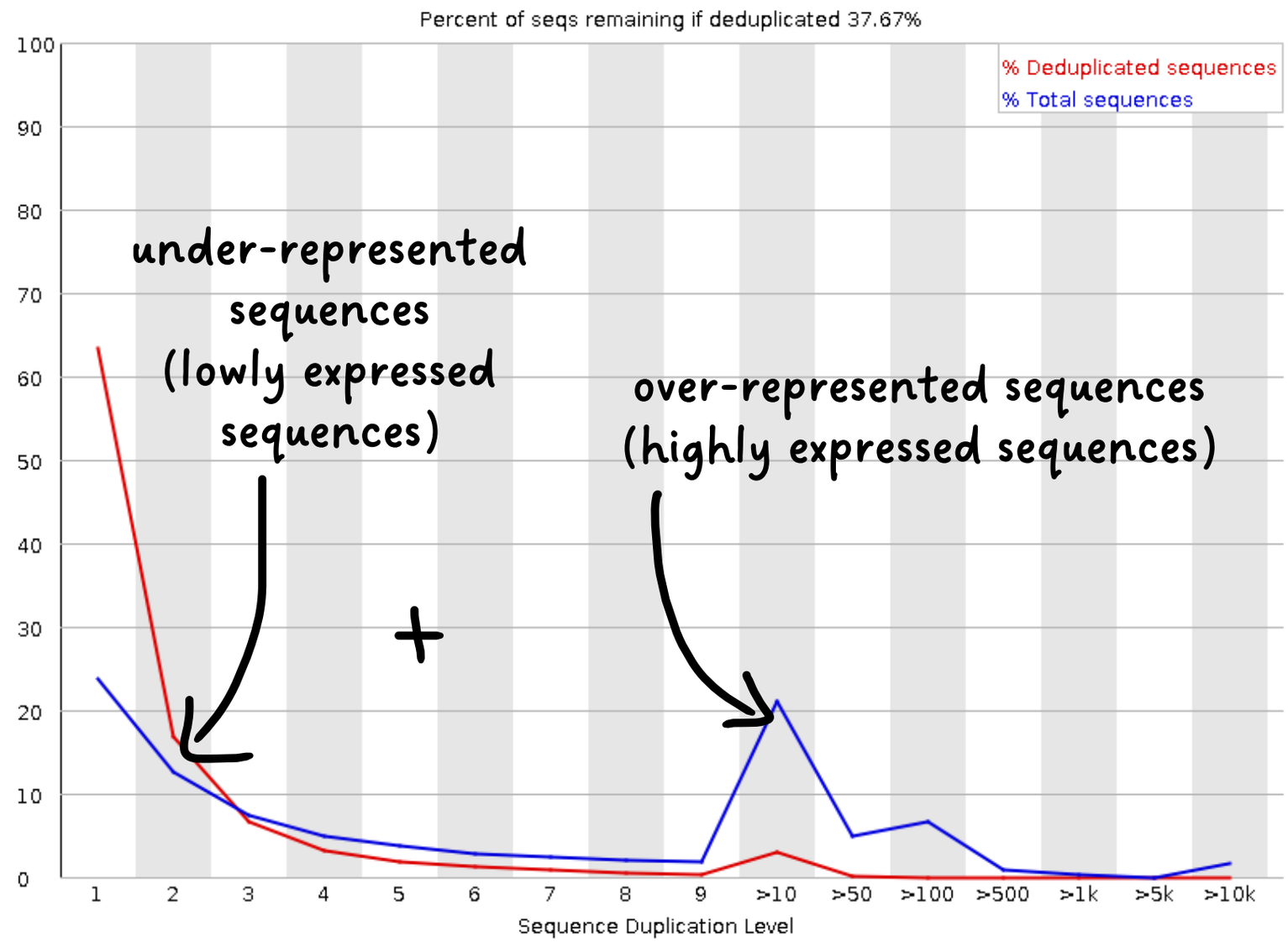
Sequence Duplication Levels



This can be due to PCR overamplification, or over-represented sequences (highly expressed sequences).
How can we tell? Look for the presence of both!
(But even then, it's really hard to know if its from over amplifying).

FastQC html report

These reports are important!

FastQC Report

Summary

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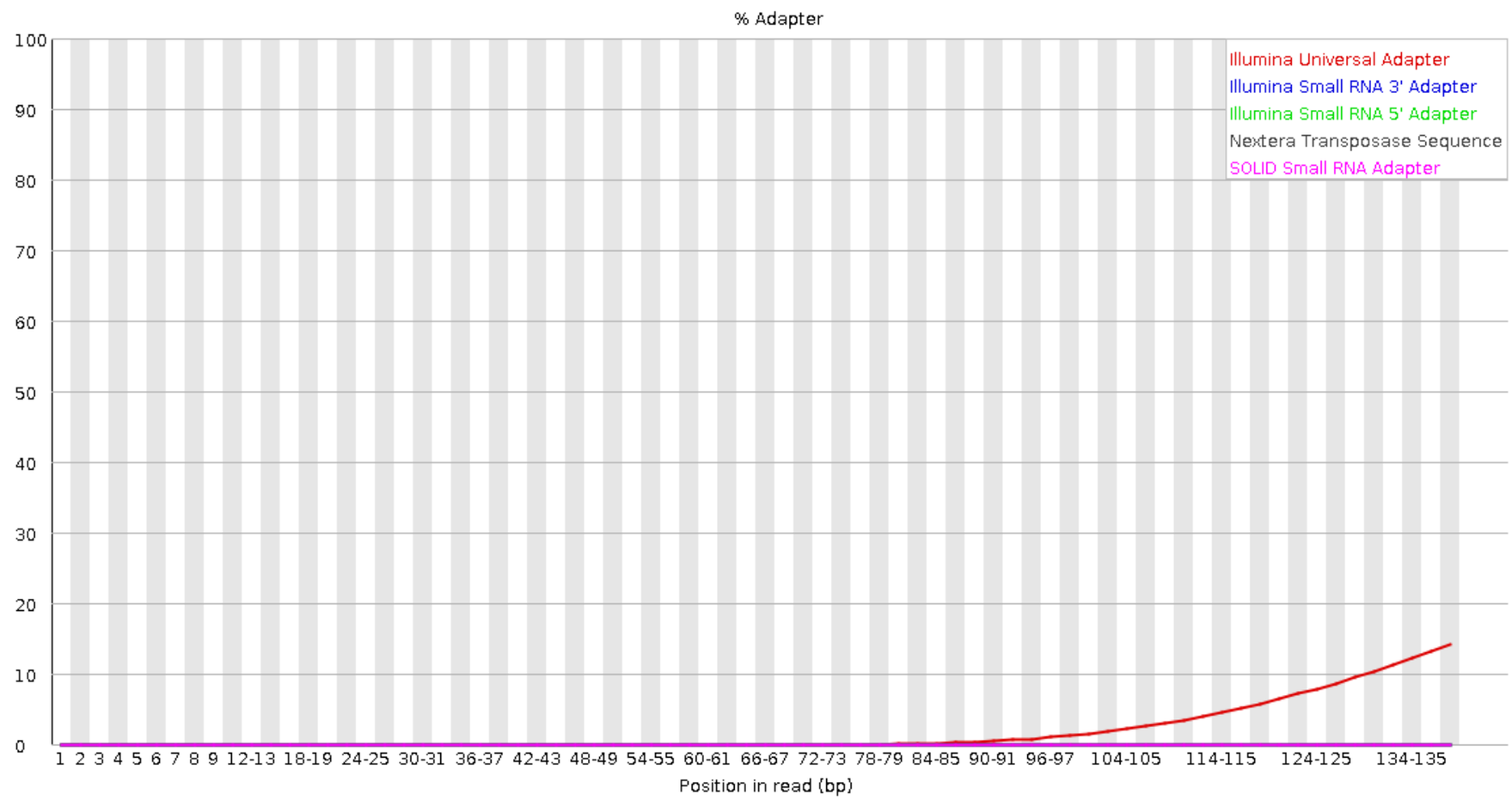
Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GATCGGAAGAGCACACGTCTGAACTCCAGTCACGTCCGCACATCTCGTAT	302627	1.7716539466510444	TruSeq Adapter, Index 18 (97% over 40bp)

Adapter Content

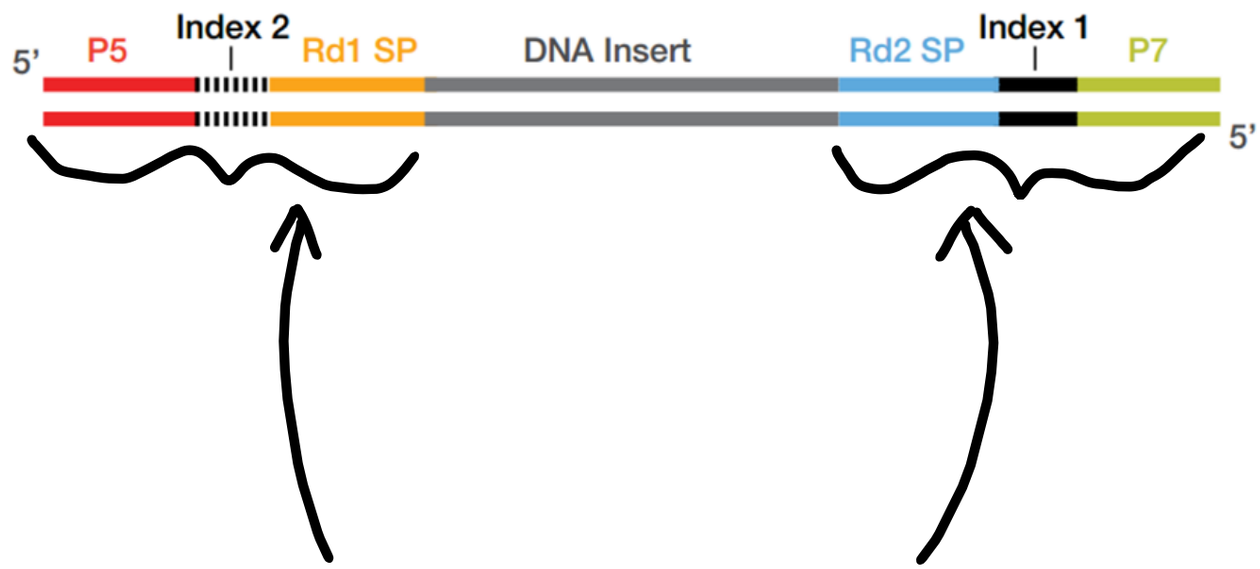


They tell us what Illumina adapters we need to get rid of!
Overrepresented sequences can also tell us if we have a lot of rRNA.

Assessing Quality & Trimming

What are the things I need to trim?

- Illumina Adapters!



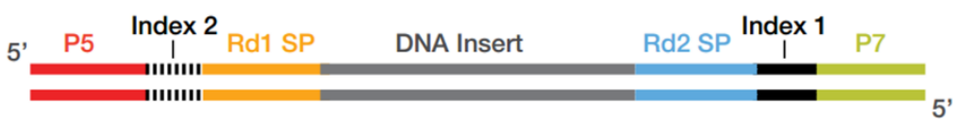
You will need to trim the adapters out of your data

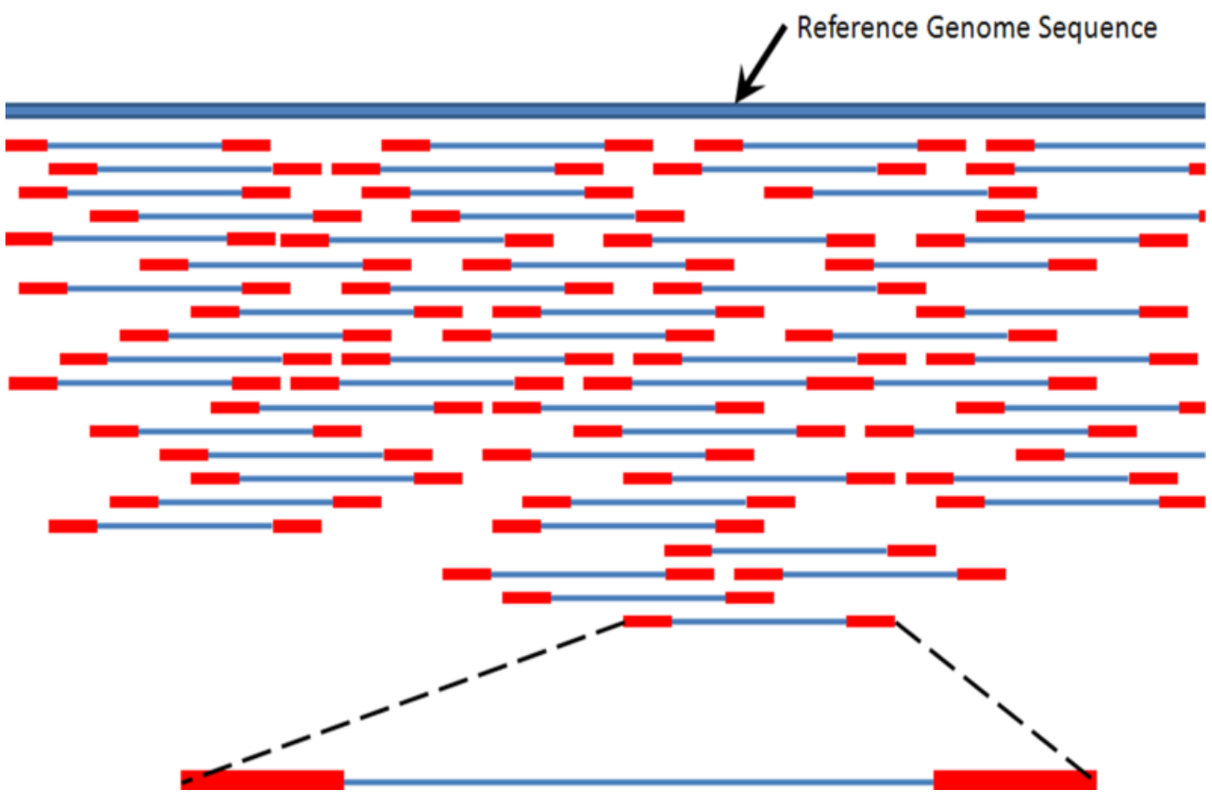
https://support.illumina.com/ko-kr/bulletins/2020/06/illumina-adapter-portfolio.html

Assessing Quality & Trimming

What are the things I need to trim?

- Illumina Adapters!





They get in the way of read mapping!

The mapping programs look for overlapping reads, and the Illumina adapters are artificial and create noise during this process.

https://support.illumina.com/ko-kr/bulletins/2020/06/illumina-adapter-portfolio.html
https://biocorecrg.github.io/RNAseq_course_2019/alignment.html

Assessing Quality & Trimming

What are the things I need to trim?

For most modern-day RNA-seq datasets, the biggest things to filter are:

- Illumina adapters
- Base quality
- Sequence quality

Assessing Quality & Trimming

What are the things I need to trim?

For most modern-day RNA-seq datasets, the biggest things to filter are:

- Illumina adapters
- Base quality
- Sequence quality

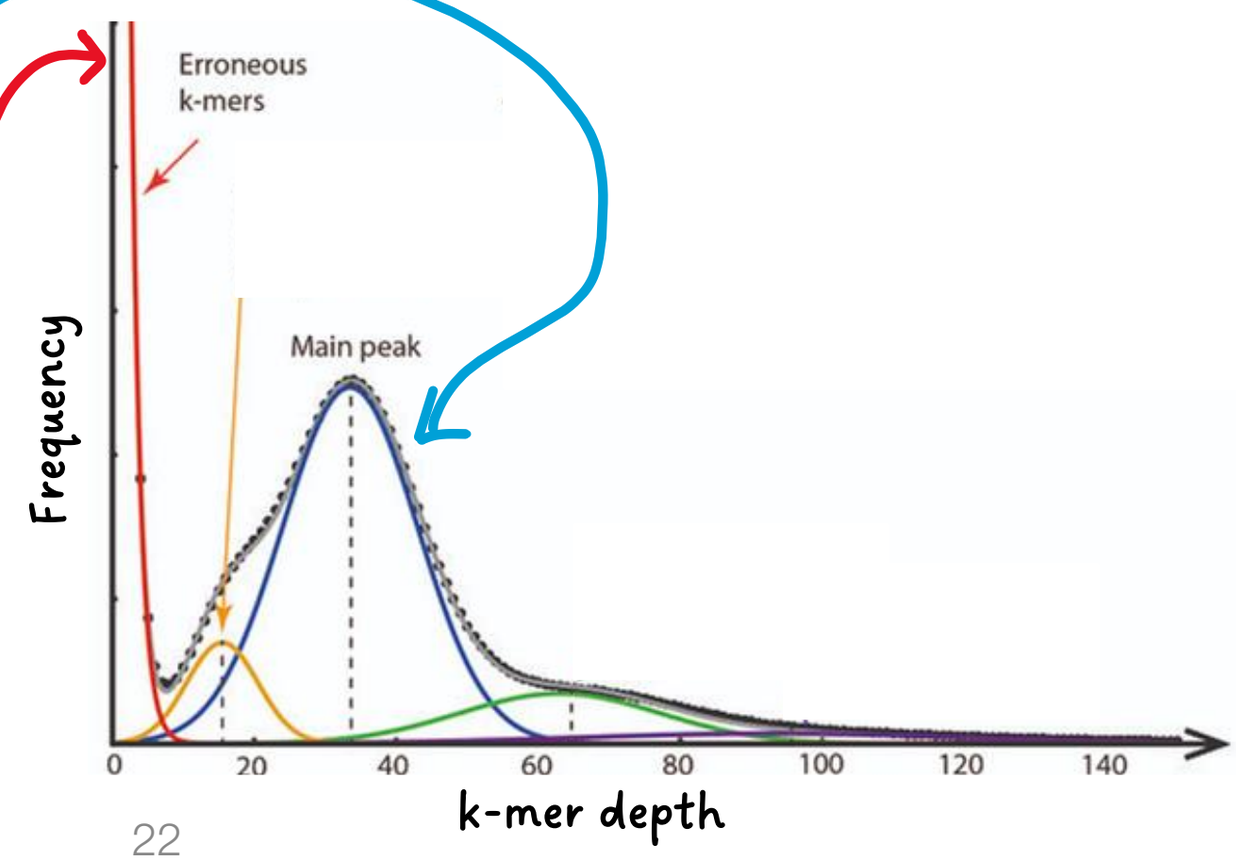
Sometimes you might hear people also filtering for:

- k-mers = reads that are broken up into small pieces (like 5-bps)

Real biological sequences produce many repeated k-mers.

Sequencing errors tend to produce rare, unique k-mers.

Filtering by k-mer coverage can remove noise, but many aligners already filter these out.



How to Trim?

Table 5.1 Read Processing Software

Software	De-mulitplexing	Adaptor Trimming	Quality Filtering/ Trimming	K-mer Filtering	K-mer Normalization
FASTX-Toolkit	✔	✔	✔		
Goby	✔	✔			
khmer				✔	✔
NGS_backbone		✔	✔		
Stacks	✔	✔	✔	✔	✔
trimmomatic		✔	✔		
biopieces	✔	✔	✔		

Which tool is the best?

How to Trim?

Table 5.1 Read Processing Software

Software	De-mulitplexing	Adaptor Trimming	Quality Filtering/ Trimming	K-mer Filtering	K-mer Normalization
FASTX-Toolkit	✔	✔	✔		
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khmer				✔	✔
NGS_backbone		✔	✔		
Stacks	✔	✔	✔	✔	✔
trimmomatic		✔	✔		
biopieces	✔	✔	✔		

Most sequencing facilities will do this for you

How to Trim?

Table 5.1 Read Processing Software

Software	De-mulitplexing	Adaptor Trimming	Quality Filtering/ Trimming	K-mer Filtering	K-mer Normalization
FASTX-Toolkit	✔	✔	✔		
Goby	✔	✔			
khmer				✔	✔
NGS_backbone		✔	✔		
Stacks	✔	✔	✔	✔	✔
trimmomatic		✔	✔		
biopieces	✔	✔	✔		

Some trimmers rely on downstream assemblers to filter and normalize by k-mers (most modern assembler do their own thing to filter k-mers!)

How to Trim?

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khmer				✔	✔
NGS_backbone		✔	✔		
Stacks	✔	✔	✔	✔	✔
trimmomatic		✔	✔		
biopieces	✔	✔	✔		

Some programs are better suited for different methods.

Stacks was primarily designed for restriction enzyme-based data, such as RAD-seq, used in population genomics.

How to Trim?

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Software	De-mulitplexing	Adaptor Trimming	Quality Filtering/ Trimming	K-mer Filtering	K-mer Normalization
FASTX-Toolkit	✔	✔	✔		
Goby	✔	✔			
khmer				✔	✔
NGS_backbone		✔	✔		
Stacks	✔	✔	✔	✔	✔
trimmomatic		✔	✔		
biopieces	✔	✔	✔		

Most important thing to focus on for most RNA-seq pipelines (ones that include alignment or assembly programs) is how fast they can trim adapters and quality filter.

How to Trim?

Table 5.1 Read Processing Software

Software	De-mulitplexing	Adaptor Trimming	Quality Filtering/ Trimming	K-mer Filtering	K-mer Normalization
FASTX-Toolkit	✔	✔	✔		
Goby	✔	✔			
khmer				✔	✔
NGS_backbone		✔	✔		
Stacks	✔	✔	✔	✔	✔
trimmomatic		✔	✔		
biopieces	✔	✔	✔		

Trimmomatic is designed to handle paired-end reads generated from Illumina sequencing and is preferably implemented in RNA seq pipelines.

Trimmomatic

Input:
- sample_R1.fastq
- sample_R2.fastq

Output:
- sample_R1_paired.fastq
- sample_R2_paired.fastq
- sample_R1_unpaired.fastq
- sample_R2_unpaired.fastq

Removes/Filters:
- Illumina adapters
- Bases with low quality
- Reads with low sequence quality
- Reads that are too short

Github: https://github.com/usadellab/Trimmomatic
Manual: usadellab.org/cms/uploads/supplementary/Trimmomatic/TrimmomaticManual_V0.32.pdf

Trimmomatic

1. Start by locating the example fastq files SRR…_1.fastq, SRR…_2.fastq (R1 = forward and R2 = reverse reads).

2. You might already have the R1 file in your folder. But let's copy the new R2 file too. So, we can delete the old and copy over the new.

3. Trimmomatic is a java program! It is not a typical compile binary, instead it runs through the Java runtime. This means we will run it through java (instead of by itself like fastqc).

4. Let's also copy over the ***Trimmomatic directory*** over to your home directory. This contains the program file and adapter sequences.

5. Let's make our first script to run Trimmomatic!

Trimmomatic

```
ubuntu@bioinformatics-1-do-not-delete:/data/workshop/Trimmomatic-0.36$ ls
LICENSE  adapters  trimmomatic-0.36.jar
```

This is the program file.

This is a folder of Illumina adapters.

```
ubuntu@bioinformatics-1-do-not-delete:/data/workshop/Trimmomatic-0.36$ ls adapters/
NexteraPE-PE.fa  TruSeq2-PE.fa  TruSeq2-SE.fa  TruSeq3-PE-2.fa  TruSeq3-PE.fa  TruSeq3-SE.fa
```

```
>PrefixPE/1
TACACTCTTTCCCTACACGACGCTCTTCCGATCT
>PrefixPE/2
GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT
>PE1
TACACTCTTTCCCTACACGACGCTCTTCCGATCT
>PE1_rc
AGATCGGAAGAGCGTCGTGTAGGGAAAGAGTGTA
>PE2
GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT
>PE2_rc
AGATCGGAAGAGCACACGTCTGAACTCCAGTCAC
```

Here is an example of the adapter sequences that Illumina uses.

We want to remove these artificial sequences from our data.

Trimmomatic

So how do we write a script?

Let's start by checking our home directory.

```
ubuntu@bioinformatics-1-do-not-delete:~$ ls
01_bashgame  01_fastqfiles  Trimmomatic-0.36  test.txt
```

I have all my fastq files in a directory called ***01_fastqfiles*** in my home.

I have the ***Trimmomatic-0.36*** directory in my home too.

Running lots of programs can get messy quickly, it's best to keep things tidy from the start!

Trimmomatic

So how do we write a script?

Let's make our first script, in our home directory.

> nano ./simple_trim.sh #This is a simple program to write a script

```
  GNU nano 6.2                                    simple_trim.sh
#!/bin/bash
mkdir -p ~/02_trimreads ~/02_trimreads/logs

java -jar /home/ubuntu/Trimmomatic-0.36/trimmomatic-0.36.jar PE \
/home/ubuntu/01_fastqfiles/SRR19736866_1.fastq /home/ubuntu/01_fastqfiles/SRR19736866_2.fastq \
/home/ubuntu/02_trimreads/SRR19736866_R1_paired.fastq /home/ubuntu/02_trimreads/SRR19736866_R1_unpaired.fastq \
/home/ubuntu/02_trimreads/SRR19736866_R2_paired.fastq /home/ubuntu/02_trimreads/SRR19736866_R2_unpaired.fastq \
ILLUMINACLIP:/home/ubuntu/Trimmomatic-0.36/adapters/TruSeq3-PE-2.fa:2:30:10:2:keepBothReads LEADING:3 TRAILING:3 MINLEN:75 \
&>> /home/ubuntu/02_trimreads/logs/Trimmomatic_SRR19736866.log
```

> screen # Let's run this script using screen
> ./simple_trim.sh

CTRL + a
Then release and press *d*

> htop # Make sure it is running, then hit "q" to quit htop

Trimmomatic

So how do we write a BETTER script?

While our first script is running, let's see if we can make improvements.

> cp simple_trim.sh advanced_trim.sh

Let's start by adding variables.

```bash
#!/bin/bash
# Advanced version with variables
# Define the location of programs and folders
TRIMMOMATIC="/home/ubuntu/Trimmomatic-0.36"
INPUT_DIR="/home/ubuntu/01_fastqfiles"
OUTPUT_DIR="/home/ubuntu/02_trimreads"
LOG_DIR="/home/ubuntu/02_trimreads/logs"

# Make these directories (-p flag indicates only do it, if these dont already exist)
mkdir -p "$OUTPUT_DIR" "$LOG_DIR"

# Here is our Trimmomatic command.
java -jar "$TRIMMOMATIC/trimmomatic-0.36.jar" PE "$INPUT_DIR/SRR19736866_1.fastq" "$INPUT_DIR/SRR19736866_2.fastq" \
  "$OUTPUT_DIR/SRR19736866_R1_paired.fastq" "$OUTPUT_DIR/SRR19736866_R1_unpaired.fastq" \
  "$OUTPUT_DIR/SRR19736866_R2_paired.fastq" "$OUTPUT_DIR/SRR19736866_R2_unpaired.fastq" \
  ILLUMINACLIP:"$TRIMMOMATIC/adapters/TruSeq3-PE-2.fa":2:30:10:2:keepBothReads LEADING:3 TRAILING:3 MINLEN:75 \
  &>> "$LOG_DIR/Trimmomatic_SRR19736866.log"
```

Then let's call these variables in our command. Variables can help keep large scripts tidy by storing information, like long paths, that can be referenced later with an abbreviation.

Trimmomatic

What are the resulting files?

```
ubuntu@bioinformatics-1-do-not-delete:~/02_trimreads$ ls
SRR19736866_R1_paired.fastq  SRR19736866_R1_unpaired.fastq  SRR19736866_R2_paired.fastq  SRR19736866_R2_unpaired.fastq  logs
```

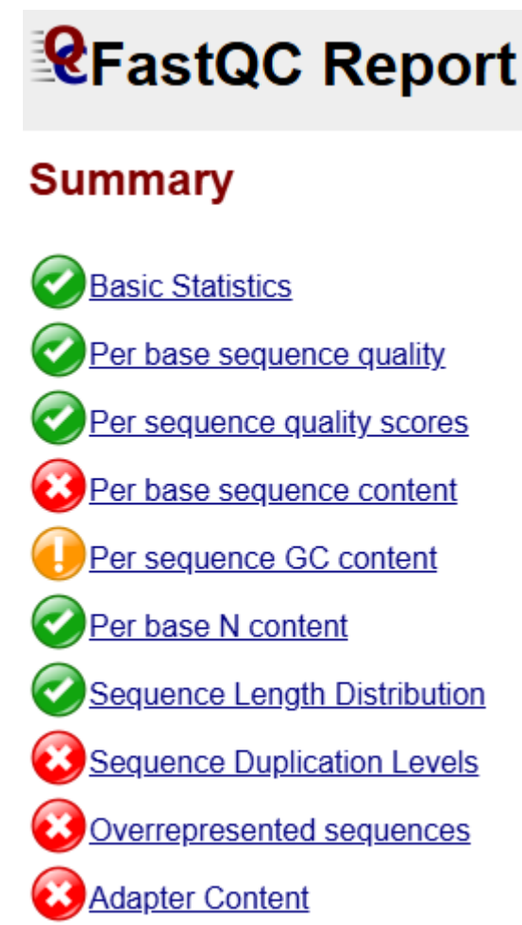
What does the log file tell us?

```
ubuntu@bioinformatics-1-do-not-delete:~/02_trimreads/logs$ cat *.log
TrimmomaticPE: Started with arguments:
 /home/ubuntu/01_fastqfiles/SRR19736866_1.fastq /home/ubuntu/01_fastqfiles/SRR19736866_2.fastq /home/ubuntu/02_trimreads/SRR197
36866_R1_paired.fastq /home/ubuntu/02_trimreads/SRR19736866_R1_unpaired.fastq /home/ubuntu/02_trimreads/SRR19736866_R2_paired.f
astq /home/ubuntu/02_trimreads/SRR19736866_R2_unpaired.fastq ILLUMINACLIP:/home/ubuntu/Trimmomatic-0.36/adapters/TruSeq3-PE-2.f
a:2:30:10:2:keepBothReads LEADING:3 TRAILING:3 MINLEN:75
Using PrefixPair: 'TACACTCTTTCCCTACACGACGCTCTTCCGATCT' and 'GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT'
Using Long Clipping Sequence: 'AGATCGGAAGAGCGTCGTGTAGGGAAAGAGTGTA'
Using Long Clipping Sequence: 'AGATCGGAAGAGCACACGTCTGAACTCCAGTCAC'
Using Long Clipping Sequence: 'GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT'
Using Long Clipping Sequence: 'TACACTCTTTCCCTACACGACGCTCTTCCGATCT'
ILLUMINACLIP: Using 1 prefix pairs, 4 forward/reverse sequences, 0 forward only sequences, 0 reverse only sequences
Quality encoding detected as phred33
Input Read Pairs: 17081609 Both Surviving: 11893076 (69.63%) Forward Only Surviving: 4562609 (26.71%) Reverse Only Surviving: 1
86114 (1.09%) Dropped: 439810 (2.57%)
TrimmomaticPE: Completed successfully
```

FastQC

Now let's see how trimming affected our quality scores

1. Start by locating your newly trimmed and paired R1 fastq file.
2. Make sure you are in the directory where the R1 file is and run FastQC in a ***screen*** session.
3. Try ***disconnecting*** from your screen session.
4. Check if your program is running using ***htop***.
5. Once done, transfer your new html file over using FileZilla.



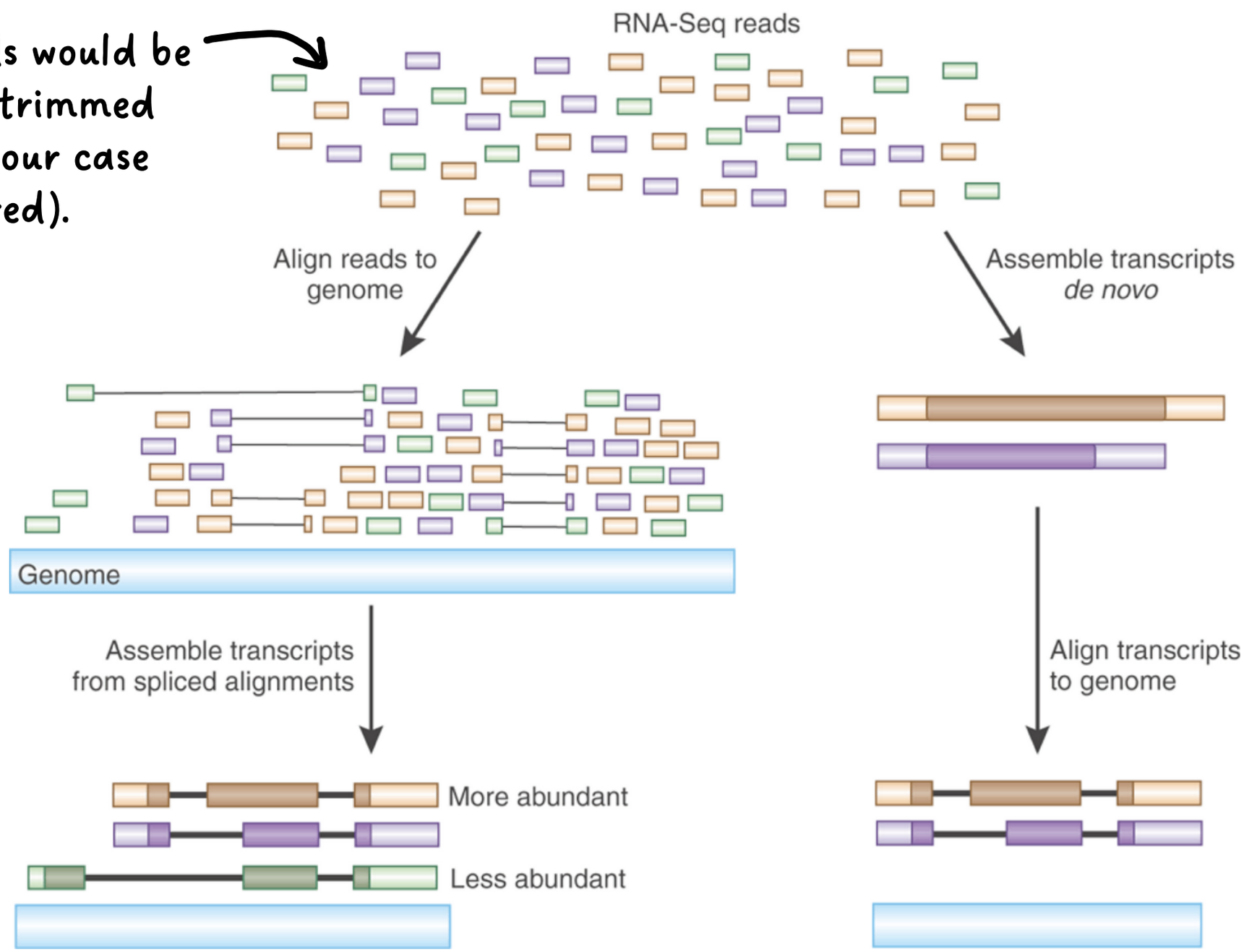
How has your quality report changed? Has it improved in the areas that were problematic?

Aligning Reads to a Genome

- What does aligning reads to a genome really mean?
- What are the different programs/tools I can use?
- What if I don't have a genome?

Align vs Assemble Reads to a Genome

These reads would be already trimmed (and in our case paired).

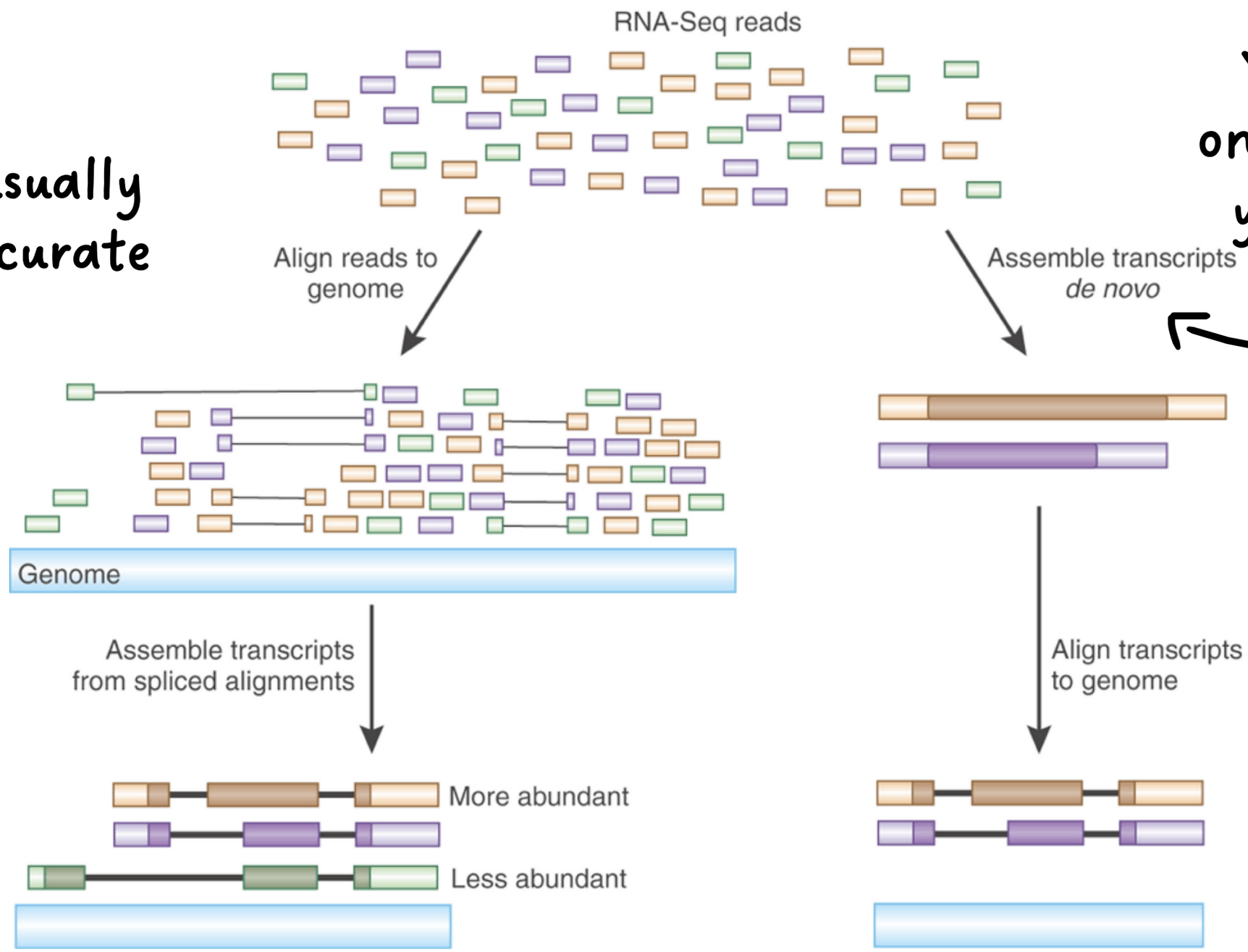


Haas & Zody 2010 Nature Biotechnology

Align vs Assemble Reads to a Genome

This is usually more accurate

You should only do this if you have to



Ex. HISAT2 (uses reference genome)
Salmon (uses reference transcriptome)

Ex. Trinity (builds its own *de novo* assembly)

Haas & Zody 2010 Nature Biotechnology

Aligning Reads to a Genome

What is actually going on?

1. We start by building an index

(You will build an index once per genome. It allows the aligner to run faster than it could if it needed to align to the whole genome.)

2. Then we align our reads to this index

(Usually aligners breaks the reads into smaller pieces, uses the index to quickly find where those segments occur in the genome, extend those matches to build a full alignment.)

Haas & Zody 2010 Nature Biotechnology

Building an Index

Let's start by building our index

1. Start by locating the genome directory.
2. Copy the genome directory over to your home directory.
3. To keep things organized, lets make 2 new directories.

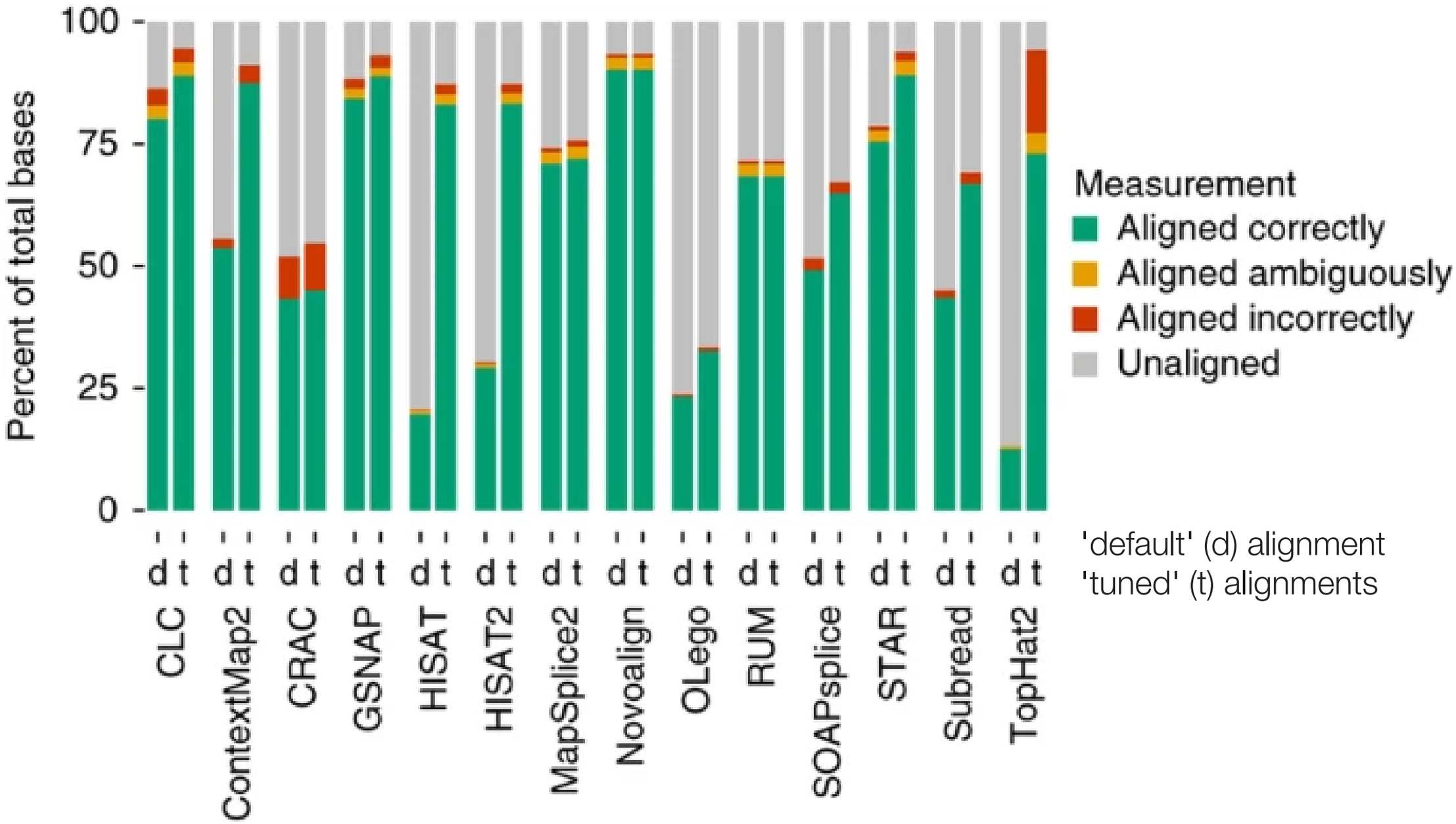
```
> mkdir -p ~/03_hisat2
> mkdir -p ~/03_hisat2/hisat2_index
```

4. Let's build our index inside of our hisat2_index directory. We only need to build one index per genome.
5. Run the hisat2-build command in a screen session.

```
> hisat2-build ~/genome/*.fna genome_index
```

(This will take some time, so we can learn more about HISAT2 while it runs.)

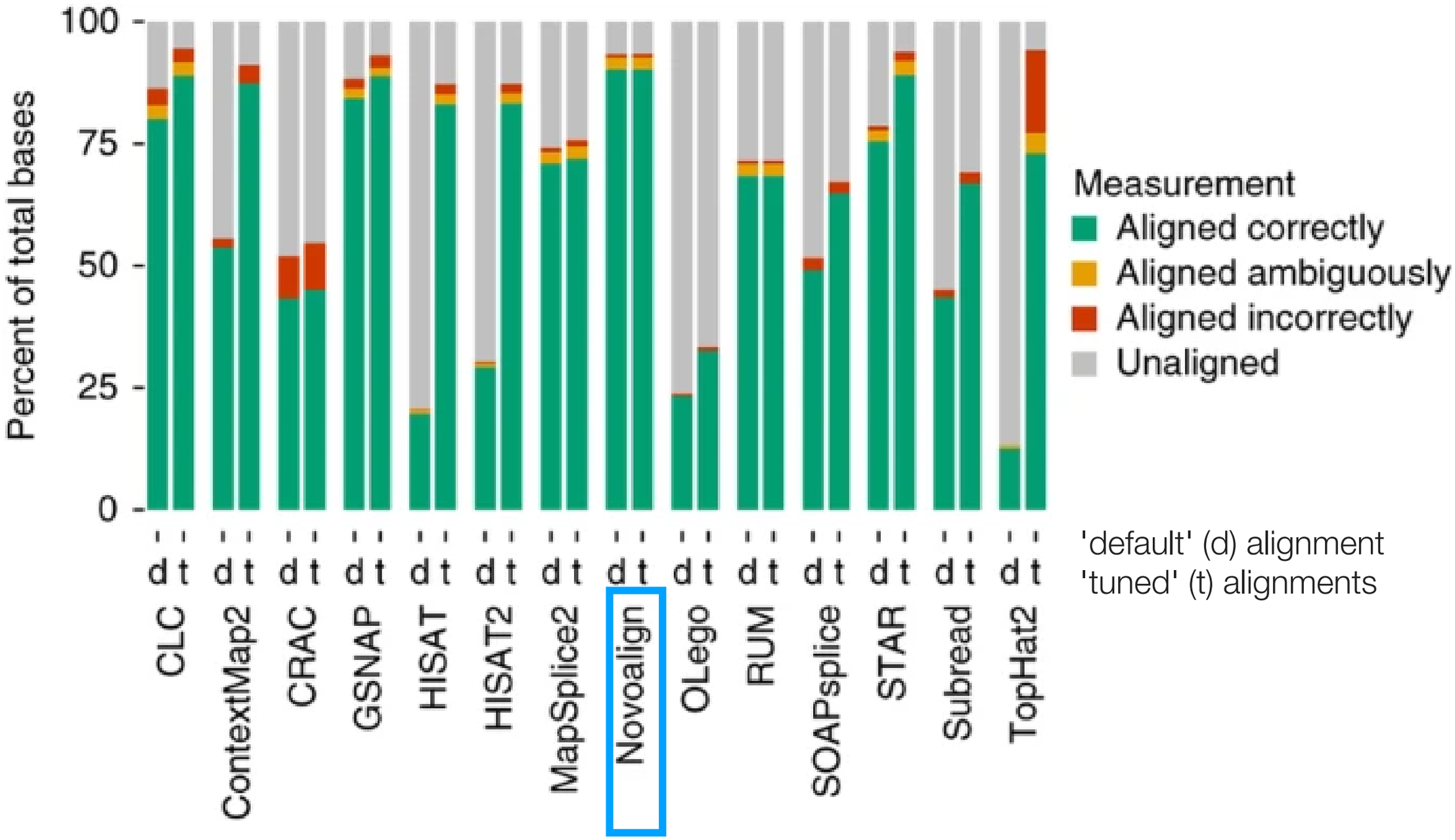
Different Alignment Tools



Figures likes these make finding the best tool seem simple, but it can be more complicated than that!

Baruzzo et al 2017

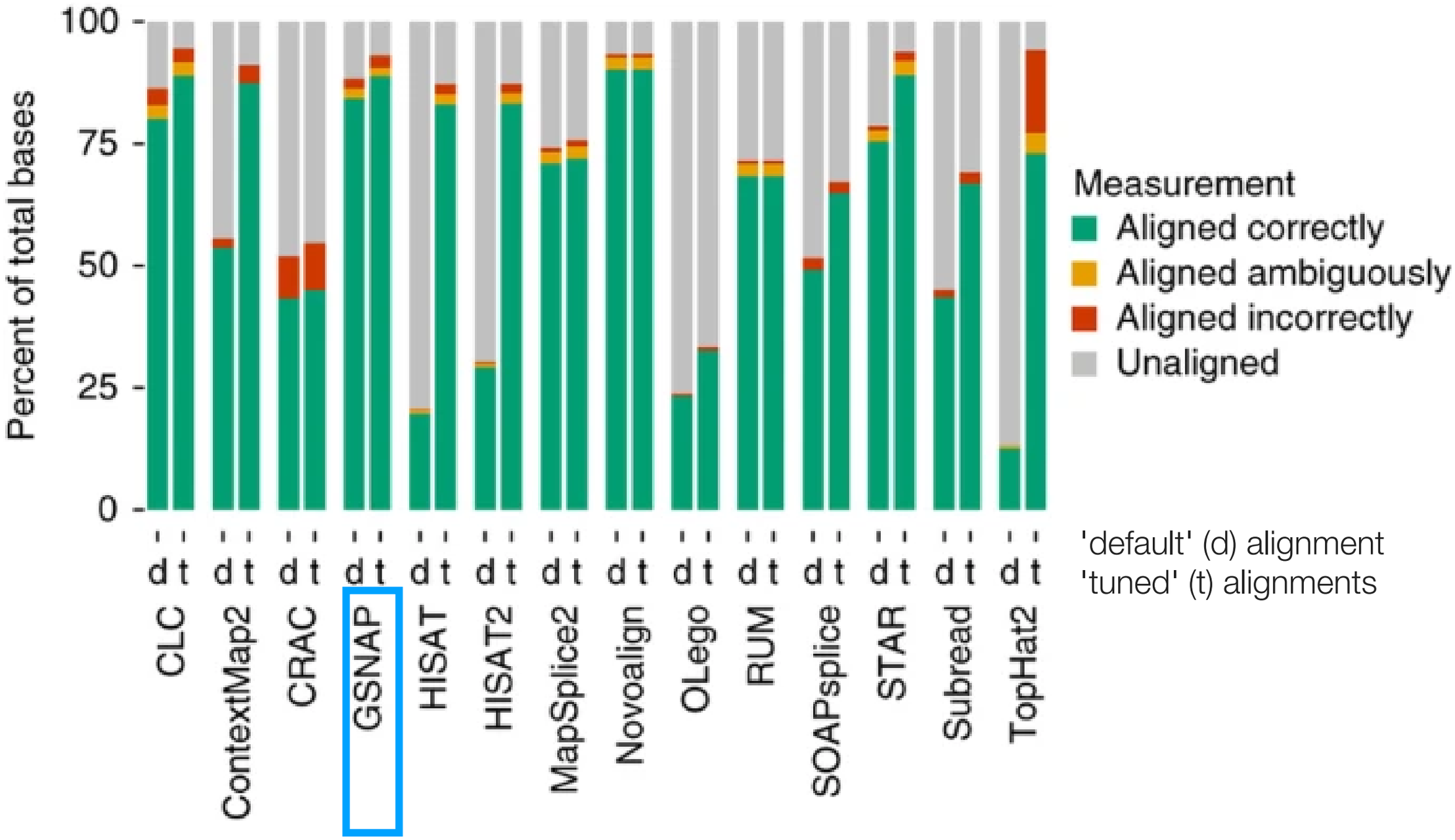
Different Alignment Tools



Great, but more than $1200/year.

Baruzzo et al 2017

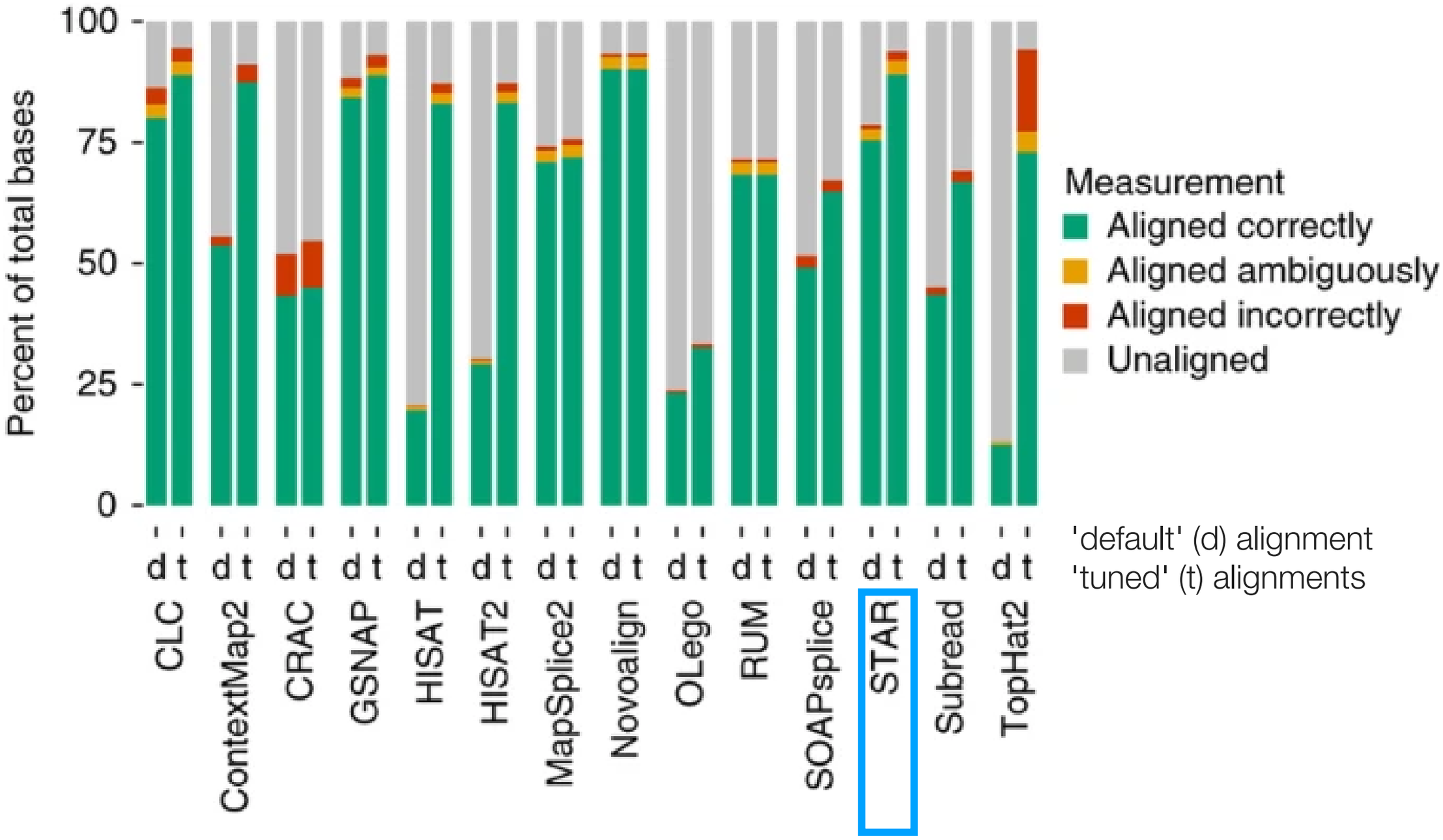
Different Alignment Tools



Really accurate, but slow! Not ideal for large datasets.

Baruzzo et al 2017

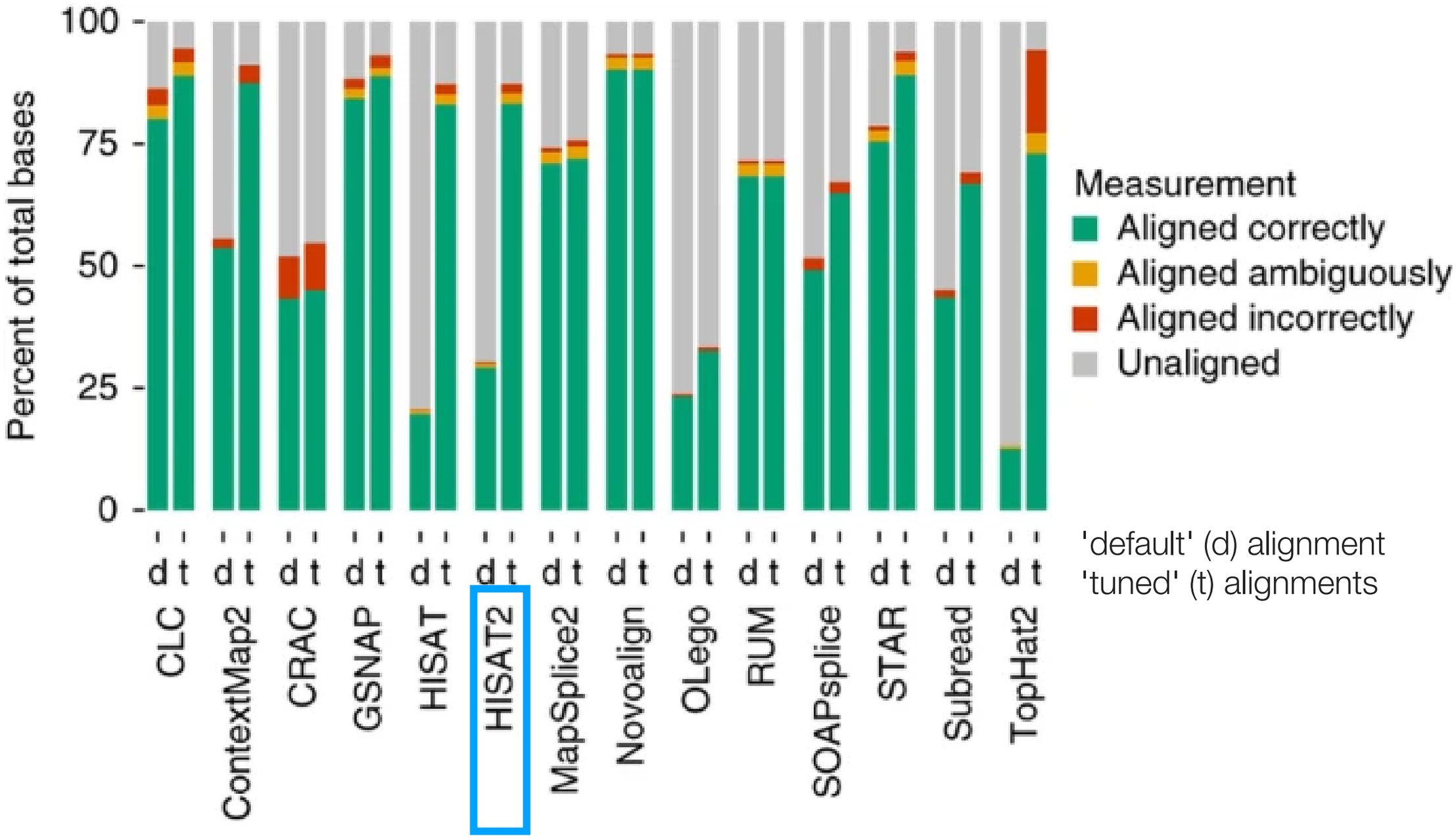
Different Alignment Tools



Accurate & fast, and good for large datasets. Handles splicing patterns well. Requires lots of RAM and a bit more difficult to setup!

Baruzzo et al 2017

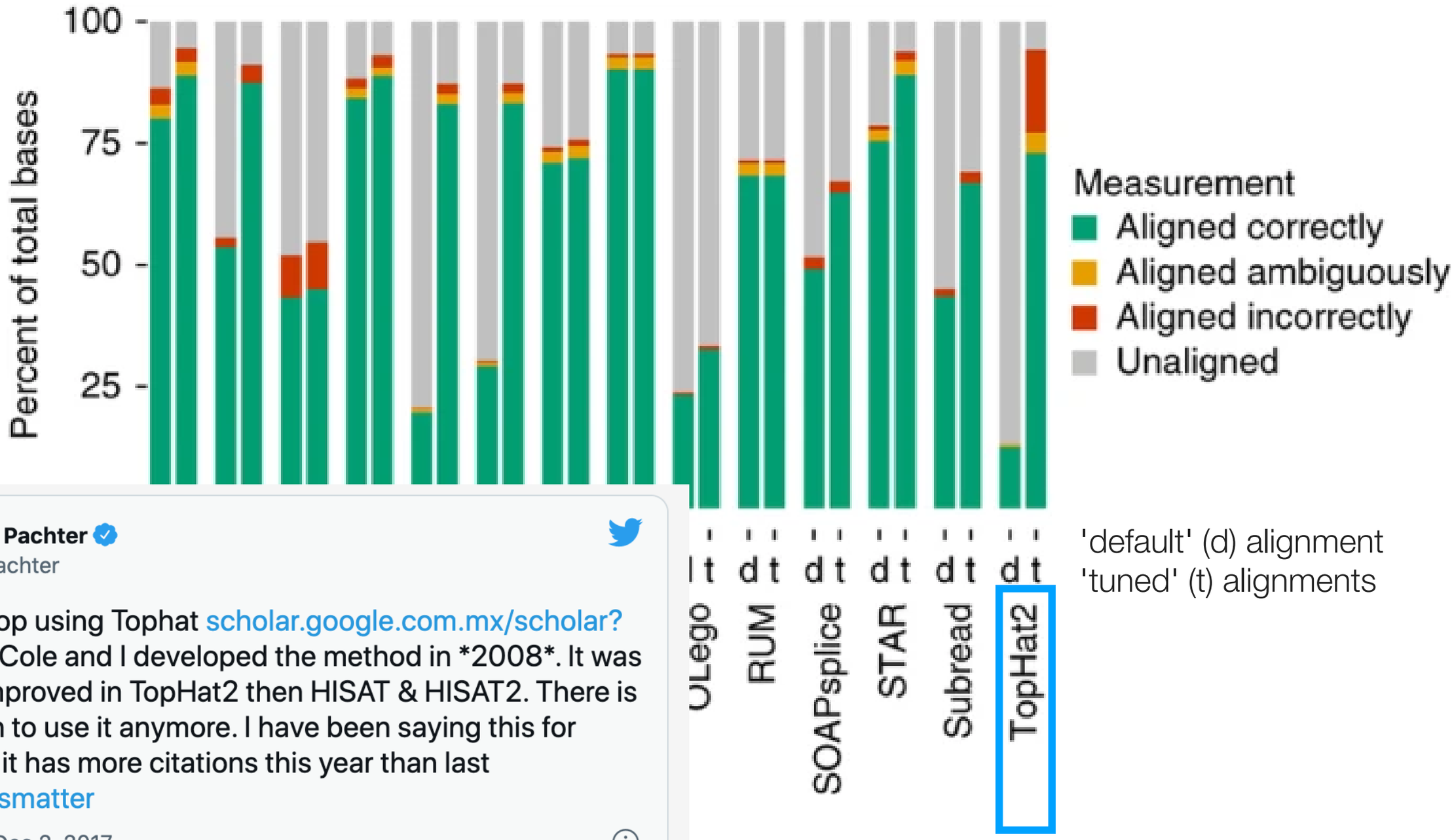
Different Alignment Tools



Accurate when parameters are tuned, but also fast and memory-efficient for large datasets. Popular implementation in analysis pipelines.

Baruzzo et al 2017

Different Alignment Tools



Baruzzo et al 2017

Running HISAT2

Let's use our index to align our reads

1. Lets re-attach our screen session and check the progress. Did everything run successfully?

2. Using our new index, lets run hisat2 on our example samples. If we had more samples, we would need to run this separately on each of our samples.

3. Let's start by creating a log directory.
   > mkdir –p ~/03_hisat2/log

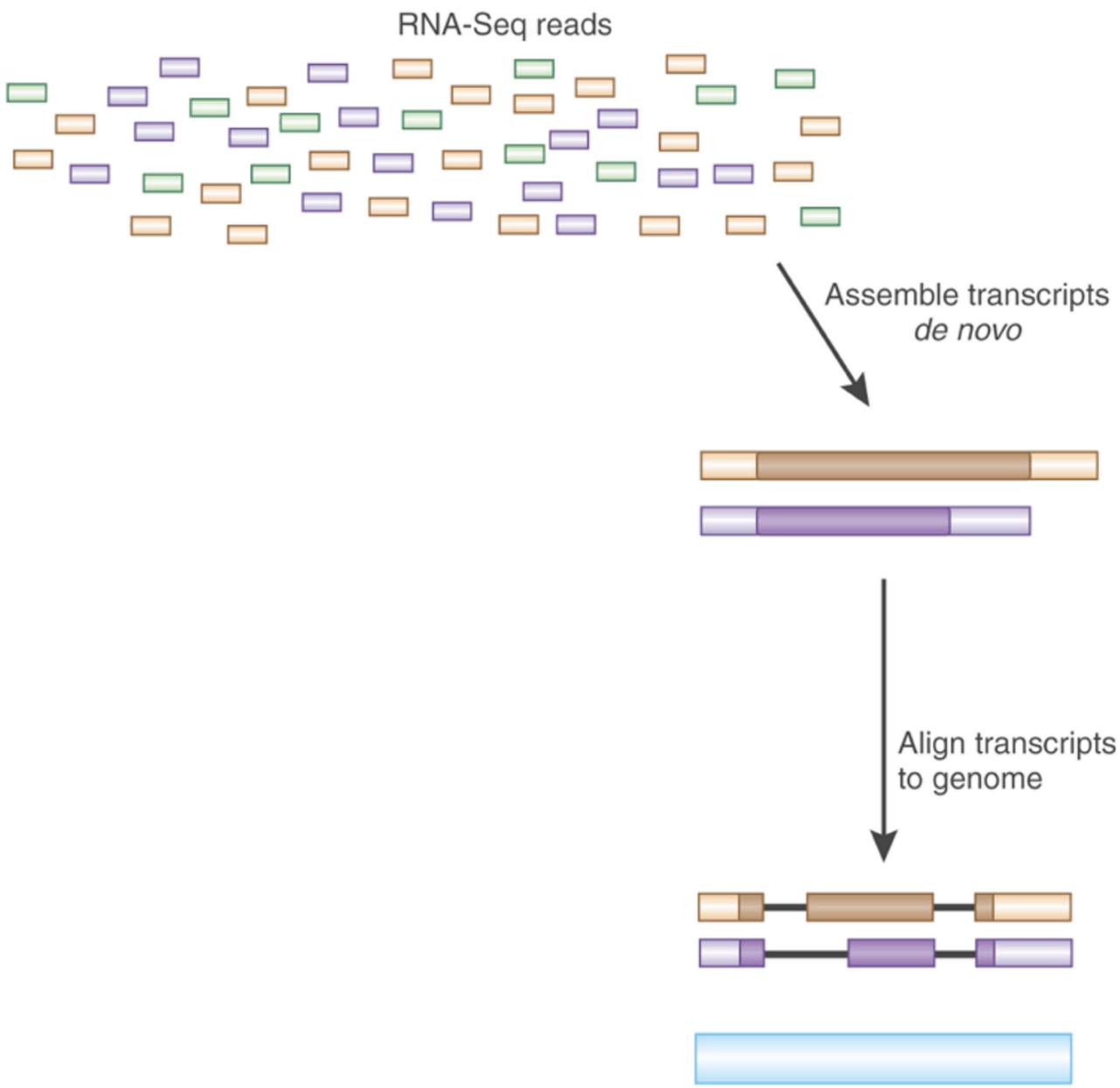
4. Now we can make a script.
   > nano run_hisat2.sh

Running HISAT2

```
  GNU nano 6.2                                   run_hisat2.sh *
#!/bin/bash
# Run HISAT2 for each sample

# Make a log directory
mkdir -p ~/03_hisat2/log

# --dta This option is needed to later run StringTie.
# -x This is the folder where you built your genome index.
# -1 This is a list of your forward reads.
# -2 These are your reverse reads.
# -S This is an output and is a SAM alignment file for each read.
# log.txt This is your alignment summary/log.
# -p This is the number of threads to run your process on.

hisat2 --dta -x ~/03_hisat2/hisat2_index/genome_index \
-1 ~/02_trimreads/SRR19736866_R1_paired.fastq \
-2 ~/02_trimreads/SRR19736866_R2_paired.fastq \
-S ~/03_hisat2/SRR19736866.sam \
-p 3 \
 &>> ~/03_hisat2/log/SRR19736866_log.txt
```

We can let this run in screen, and check the output file next session!

What if I don't have a genome?

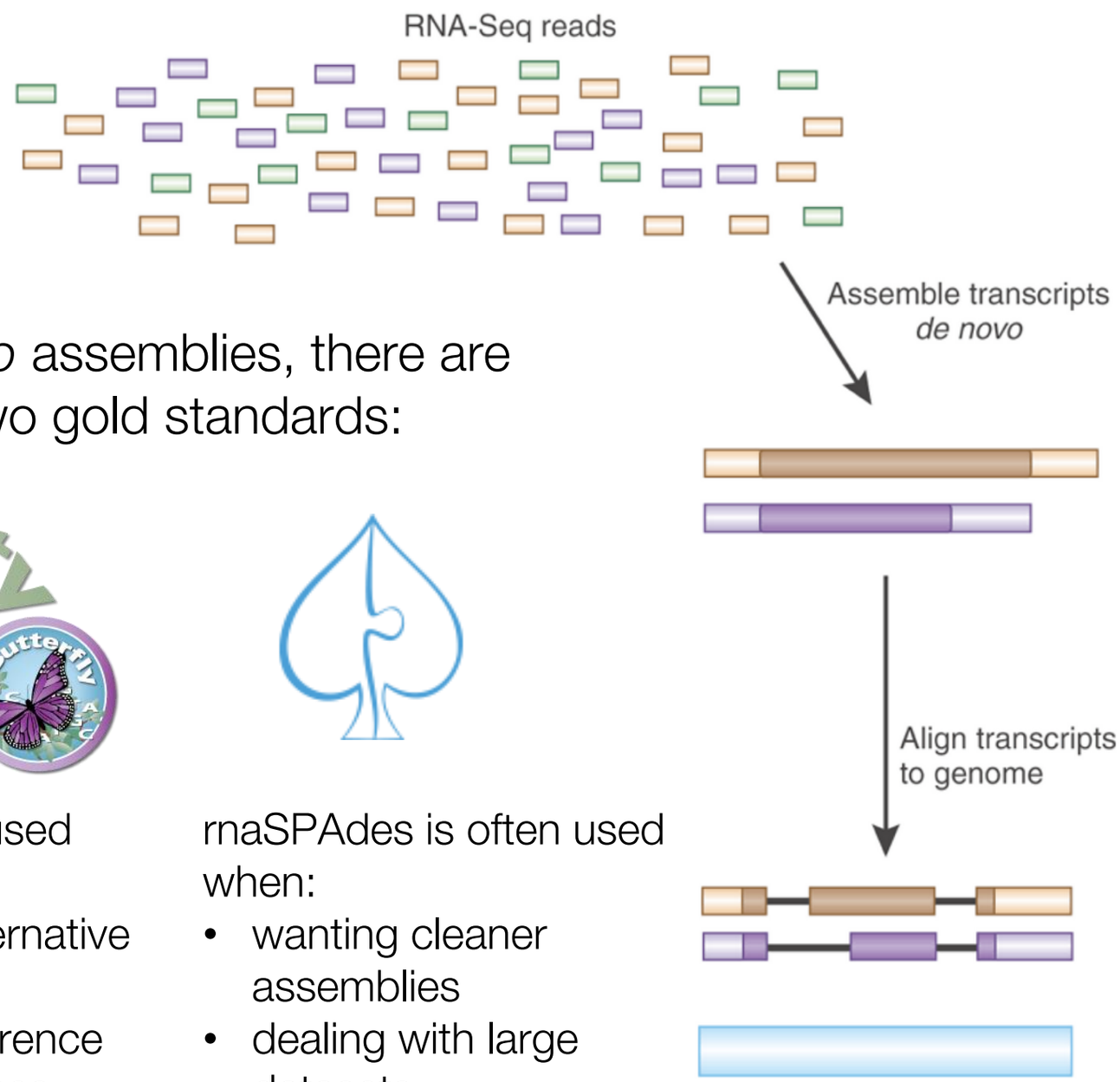


You should only do this if you have to

-needs lots of RAM
-inflates unique transcript counts
-more susceptible to including contamination or sequencing errors

Haas & Zody 2010 Nature Biotechnology

What if I don't have a genome?



You should only do this if you have to

-needs lots of RAM
-inflates unique transcript counts
-more susceptible to including contamination or sequencing errors

For *de novo* assemblies, there are currently two gold standards:

Trinity is often used when:

- studying alternative splicing
- building reference transcriptomes
- maximizing transcript discovery

rnaSPAdes is often used when:

- wanting cleaner assemblies
- dealing with large datasets
- reducing redundant transcripts

https://github.com/trinityrnaseq/trinityrnaseq/wiki
https://ablab.github.io/spades/rna.html

Haas & Zody 2010 Nature Biotechnology

Running Trinity

How is it similar to running HISAT2?

1. It's a simple command, also requiring the forward (left/R1) and reverse (right/R2) files.

```
Trinity --seqType fq --left reads_1.fq --right reads_2.fq --CPU 6 --max_memory 20G
```

Running Trinity

How is it different to running HISAT2?

1. Trinity has many dependencies (or other programs it uses when it runs). This can make it difficult to install, but also useful in other aspects (for example, you can run Trimmomatic in Trinity).

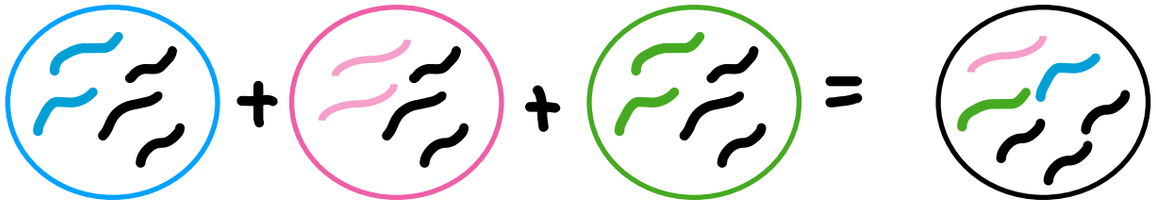
2. In HISAT2, we run samples one at a time, because we already have a reference genome; however, in Trinity you would want to include all your samples (for the same assemblage) in a single run.

```
./Trinity --trimmomatic --seqType fq --max_memory 120G \
--left sample1_1.fastq, sample2_1.fastq,... sample10_1.fastq \
--right sample1_2.fastq, sample2_2.fastq,...sample10_2.fastq \
--CPU 6
```

Running Trinity

How is it different to running HISAT2?

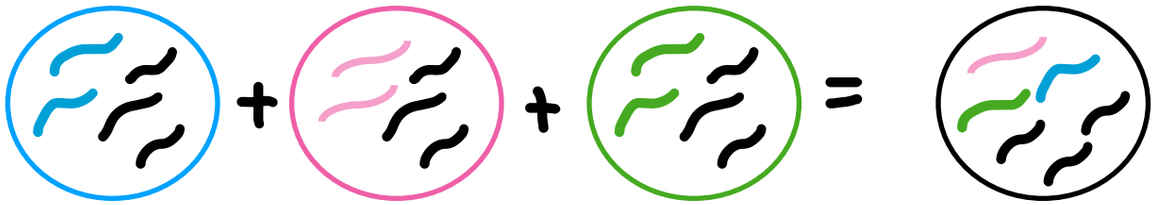
3. Since Trinity is building a reference, it works best when you have many different biological replicates to generate the reference transcriptome.



Running Trinity

How is it different to running HISAT2?

3. Since Trinity is building a reference, it works best when you have many different biological replicates to generate the reference transcriptome.



4. If you suspect differential isoform usage, it might be best to separate these samples into different assemblages (such as tissue- or development-specific assemblies).

   - Highly expressed transcripts from one tissue may dominate graph construction and rare isoforms from the other tissue may be underrepresented.

   - The assembly graph becomes more complex because it contains many alternative paths from multiple expression regimes (leads to incorrect transcript reconstruction, fragmented assemblies, missing isoforms, longer run times).

   - However, separate assemblies require merging and redundancy removal (BLAST, or EvidentialGene).

Different Tools

A consensus has not yet been reached about the optimal approach, in practice what you do will likely be informed by the data you have.

Explore bioinformatic tools (including mappers):
https://australianbiocommons.github.io/toolfinder/

How to install and run popular RNA-seq analysis programs (including HISAT2 and Trinity):
https://bernadettebiology.weebly.com/protocols-tutorials--courses.html

Course Info

- Course Materials on Github: https://github.com/BernadetteBiology
- Photos, unless otherwise noted, are under © Bernadette J, some rights reserved (CC-BY-NC)